

HelixTerminator — Core Architecture Specification

Document: 01_core_architecture.md **Version:** V1 · Revision 1 **Go Module:** helixterm.io/core **Spec Owner:** HelixDevelopment **Date:** 2026-06-28 **Status:** DRAFT — Engineering Review **Constitution Compliance:** HelixConstitution (pinned e6504c2, helixcode-v1.1.0 line) · §11.4, §11.4.73–78

Inheritance: This document and the project that contains it extend the Helix Universal Constitution at constitution/Constitution.md. All clauses there apply unless explicitly overridden. See constitution/CLAUDE.md and constitution/AGENTS.md.

NOTE (CD-2 — flagged, not mass-rewritten): per CANONICAL_FACTS.md, the primary org/domain identity going forward is HelixDevelopment / helixterminator.io (org identity above is already correct). This document's Go module namespace and every helixterm.io host/URL/SPIFFE-ID reference throughout §1–§10 predate that domain decision and are **not** rewritten here — a domain-wide rename is a high-churn, cross-cutting change (100+ refs) tracked as DEFERRED cross-doc work in CANONICAL_FACTS.md, consistent with how 10_submodule_integration.md treats the same identity fields. Do not treat helixterm.io below as newly-canonical.

Table of Contents

1. Executive Summary & Vision
2. Complete System Architecture
3. Complete Microservices Catalog
4. Database Architecture
5. Kafka & RabbitMQ Architecture
6. Zero Trust Security Architecture
7. API Specification
8. Performance Architecture

1. Executive Summary & Vision

1.1 Mission Statement

HelixTerminator is a next-generation, enterprise-grade SSH client platform designed from first principles to replace and decisively surpass Termius across every dimension: security posture, scalability, developer experience, collaboration depth, extensibility, and operational intelligence. Where Termius is a well-polished productivity tool for individual operators and small teams, HelixTerminator is a full-stack, microservices-native platform targeting hyperscale engineering organizations, regulated-industry enterprises, managed service providers, and cloud-native infrastructure teams who demand audit-trail completeness, zero-trust identity enforcement, AI-augmented workflows, and continuous compliance verification baked into every layer of the stack.

HelixTerminator ships as a unified platform — a set of independently deployable, loosely coupled Go microservices backed by Apache Kafka for event streaming, RabbitMQ for command routing, PostgreSQL for durable relational state, and Redis for high-throughput caching. All clients — Web SPA, Desktop (macOS, Windows, Linux), iOS, and Android — are built on Flutter/Dart and consume a single versioned REST/WebSocket/gRPC surface.

The platform is built around five irrevocable principles:

1. **Security is not a feature — it is the substrate.** Every connection, every byte of stored credential, every API call is cryptographically authenticated. There are no unencrypted paths.
2. **Audit completeness is non-negotiable.** Every terminal command, every file transfer, every authentication event, every configuration change is persisted as an immutable, cryptographically chained audit event, available for real-time streaming and retrospective forensic analysis.
3. **Collaboration is a first-class primitive.** Multiple operators can share a terminal session with role-differentiated capabilities (read-

only observer, co-pilot, session owner) in real time, with sub-100ms propagation latency.

4. **AI augmentation is integrated, not bolted on.** The AI/Autocomplete service analyses shell history, infrastructure topology, and contextual signals to suggest commands, detect anomalies, generate runbooks, and explain terminal output — all without transmitting sensitive credential material to external model providers.
5. **HelixConstitution compliance is mechanically enforced.** Every CI/CD pipeline gate, every agent rule file, every submodule dependency graph, and every code generation pass respects the universal constitution inherited from `HelixDevelopment/HelixConstitution` (pinned e6504c2, helixcode-v1.1.0 line).

1.2 How HelixTerminator Surpasses Termius at Every Dimension

1.2.1 Credential Management

Termius: Credentials (passwords, private keys) are stored in a proprietary encrypted vault synced across devices. The encryption key derivation is opaque to the operator, and there is no verifiable zero-knowledge guarantee. The vault is a client-side construct with no server-side auditability.

HelixTerminator: The Vault Service (helixterm.io/services/vault) implements a verifiable zero-knowledge architecture. Vault items are encrypted client-side using AES-256-GCM with keys derived via Argon2id (time_cost=3, memory_cost=65536, parallelism=4). The server stores only ciphertext. Key material is never transmitted in plaintext. SSH private keys are additionally wrapped by the Certificate Authority Service (helixterm.io/services/pki) which issues short-lived signed SSH certificates (TTL=8h by default, configurable per org) so that static private keys are never stored on jump hosts. The Keychain Service (helixterm.io/services/keychain) provides hardware-backed key storage on platforms that support it (macOS Secure Enclave, Android Keystore, Windows DPAPI), using `digital.vasic.security` for the underlying AES and storage primitives.

1.2.2 Team Collaboration

Termius: Teams can share host configurations and credential groups, but there is no real-time collaborative terminal session. Screen sharing requires external tools.

HelixTerminator: The Collaboration Service (helixterm.io/services/collab) and Terminal Session Service (helixterm.io/services/terminal) jointly implement WebSocket-based real-time session sharing with the following capabilities: - **Session owner:** Full interactive control. - **Co-pilot:** Interactive input, must be granted by owner. - **Observer:** Read-only streaming with zero write access enforced at the SSH Proxy level, not just the UI. - **Synchronized scrollbar buffer:** All participants see an identical terminal state, including historical output, delivered via CRDT-based buffer synchronization over Kafka event streams. - **Broadcast sessions:** One operator can stream a terminal session to thousands of observers for training, incident response, and runbook execution demonstrations.

1.2.3 Session Recording and Audit

Termius: No built-in session recording. No audit trail. No compliance reporting.

HelixTerminator: The Session Recording Service (helixterm.io/services/recording) records every terminal session as a full-fidelity asciinema-format stream, timestamped at millisecond granularity, stored in object storage (S3-compatible), and indexed in PostgreSQL for full-text search. Recordings are cryptographically signed (Ed25519) so tampering is detectable. The Audit Service (helixterm.io/services/audit) maintains an append-only, Merkle-chained audit log of every authentication event, command execution, configuration change, and privilege escalation, compliant with SOC 2 Type II, ISO 27001, and FedRAMP Moderate control requirements. Audit events are emitted to Kafka topic `helix.audit.events` with retention of 365 days.

1.2.4 Authentication and MFA

Termius: Username/password + TOTP-based 2FA. No hardware key support beyond basic TOTP. No SSO for free/personal tiers. No certificate-based authentication beyond private key support.

HelixTerminator: Full FIDO2/WebAuthn MFA with hardware security key support (YubiKey 5, Google Titan, any CTAP2-compliant device). Biometric authentication on mobile and desktop via platform authenticator APIs. Enterprise SSO via SAML 2.0 and OIDC (Auth Service `helixterm.io/services/auth` consumes `digital.vasic.auth`). Device certificates issued at registration time, short-lived (TTL=24h renewable), providing a passwordless authentication path for CI/CD agents and automation pipelines. Passkey support (discoverable credentials) for next-generation password-free human logins.

1.2.5 AI and Automation

Termius: No AI features as of 2026. Terminal experience is purely reactive.

HelixTerminator: The AI/Autocomplete Service (`helixterm.io/services/ai`) provides:

- **Command autocomplete:** Trained on the operator's own shell history, infrastructure topology, and a corpus of DevOps best practices. Suggestions are generated locally (on-device model) or via a private inference endpoint — no data leaves the organization's trust boundary by default.
- **Anomaly detection:** Real-time analysis of command streams for suspicious patterns (bulk `rm -rf`, unexpected sudo escalations, data exfiltration signatures). Anomalies emit events to `helix.security.anomalies` Kafka topic, triggering immediate alerts and optionally session termination.
- **Runbook generation:** Given a description of an operational task, the AI service generates a multi-step runbook with inline shell commands, dependency checks, and rollback steps.
- **Terminal output explanation:** The operator can highlight terminal output and request an explanation, which is annotated inline in the terminal scrollbar buffer.
- **Incident response assist:** During a declared incident (triggered via HelixTrack integration), the AI service surfaces relevant runbooks, recent command history across team members, and infrastructure state diffs.

1.2.6 Container and Kubernetes Integration

Termius: Basic SSH connectivity. No native container or Kubernetes awareness.

HelixTerminator: The Container Registry Bridge (`helixterm.io/services/container-bridge`) and SSH Proxy Service (`helixterm.io/services/ssh-proxy`) integrate with `digital.vasic.containers` to provide:

- **Pod exec:** Direct `kubectl exec`-style terminal sessions into

Kubernetes pods via the platform, with full audit trail and recording. - **Container shell:** Docker/Podman container shell sessions without requiring an SSH daemon inside the container. - **Health dashboard:** Real-time pod health, resource utilization, and event streams sourced from the Kubernetes API, displayed in the HelixTerminator sidebar. - **Container lifecycle management:** Start, stop, restart, and tail logs for containers managed via the platform.

1.2.7 Port Forwarding and Tunneling

Terminus: Basic local/remote/dynamic port forwarding. No management plane for active tunnels.

HelixTerminator: The Port Forwarding Service (helixterm.io/services/port-forward) maintains a persistent catalog of forwarding rules (stored in PostgreSQL), manages their lifecycle (auto-reconnect with exponential backoff using `digital.vasic.recovery`), and exposes them via a management API. Local, remote, and dynamic (SOCKS5) modes. Reverse tunnels for exposing internal services to authorized external users. Tunnel metrics (bytes transferred, connection latency, error rates) streamed to `helix.metrics.tunnels` Kafka topic for Prometheus consumption.

1.2.8 SFTP and File Management

Terminus: SFTP file browser with basic upload/download. No bulk operations, no directory synchronization, no diff view.

HelixTerminator: The SFTP Service (helixterm.io/services/sftp) implements a full-featured file management experience: - **Multi-pane browser:** Two-pane local/remote file manager with drag-and-drop, bulk operations, and progress tracking. - **Directory sync:** Rsync-style synchronization between local and remote directories with diff preview and conflict resolution. - **File diff viewer:** Inline diff between local and remote versions of text files. - **Transfer queue:** Persistent transfer queue with pause/resume, retry, and priority ordering. - **Checksums:** Automatic SHA-256 checksum verification after every transfer. - **Audit integration:** Every file transfer (upload/download/delete) emits an audit event.

1.3 Target Audience

1.3.1 Individual Developers

Individual developers gain a best-in-class SSH client with intelligent autocomplete, a beautiful cross-platform Flutter UI, zero-friction host management, and AI-powered troubleshooting — all on a free tier with no degradation of core functionality.

1.3.2 Engineering Teams (5–200 engineers)

Teams gain shared host inventories, shared snippet libraries, collaborative terminal sessions, and a centralized secrets vault with per-user access control. Integration with HelixTrack (helixterm.io/services/helixtrack-bridge) links terminal sessions to open issues and sprints, enabling traceability from deployment commands to engineering work items.

1.3.3 Enterprise Organizations

Enterprises gain SOC 2 Type II-aligned audit trails, SAML 2.0/OIDC SSO, SCIM directory synchronization, role-based access control at the host/group/command level, session recording with tamper-evident storage, FIDO2 MFA enforcement, and a comprehensive compliance dashboard. The Organization Service (helixterm.io/services/org) models multi-tenant hierarchies (Organization → Teams → Members) with fine-grained permission inheritance.

1.3.4 DevOps Engineers and SREs

Seamless Kubernetes and container integration, port forwarding lifecycle management, real-time infrastructure health dashboards, runbook library with execution tracking, and AI-powered anomaly detection during incident response.

1.3.5 Managed Service Providers (MSPs)

Multi-tenant organization management, per-customer credential isolation, white-label deployment capability, comprehensive per-customer audit reporting, and customer-specific rate limit and resource quota configuration.

1.4 Bleeding-Edge Innovations

1. **Cryptographically chained audit log (Merkle structure):** Each audit event references the SHA-256 hash of the previous event, making retroactive tampering computationally infeasible and detectable by any holder of the chain.
2. **Short-lived SSH certificates:** Instead of static private keys, HelixTerminator's PKI service issues X.509-compatible SSH certificates with TTLs as short as one hour, eliminating the long-lived credential risk that underlies most SSH-related breaches.
3. **CRDT-based collaborative terminal:** Using a Conflict-free Replicated Data Type for terminal buffer state ensures that multiple collaborators always see an eventually consistent, conflict-free view of terminal output, even under high-latency network conditions.
4. **Zero-knowledge vault:** The server is architecturally incapable of reading vault contents. Client-side encryption is enforced at the protocol level, not merely by policy.
5. **On-device AI inference:** Command suggestion models are small enough (< 50MB quantized) to run entirely on-device via ONNX Runtime, eliminating the need to transmit command history to external inference endpoints.
6. **digital.vasic.docs_chain integration:** The Salsa-inspired DAG-based document dependency engine propagates specification changes across the entire documentation corpus automatically, ensuring that changes to this core architecture document trigger validation gates on all dependent specification documents.
7. **Container-native SSH sessions via digital.vasic.containers:** SSH sessions can target not just traditional SSH daemons but any container runtime — Docker, Podman, or Kubernetes pod — via the containers submodule's unified ContainerRuntime abstraction, without installing an SSH server inside the container.
8. **HelixConstitution-enforced CI/CD:** Every pull request gate verifies constitution inheritance integrity (anti-bluff audit, §11.4 anchor checks), submodule catalogue consistency, and containers-submodule mandate compliance before merge.

1.5 Market Positioning

HelixTerminator enters a market segment currently occupied by Termius (consumer/SMB SSH client), Teleport (access management platform), and HashiCorp Boundary (zero-trust access). HelixTerminator's differentiation is the synthesis of all three into a

single, developer-first platform: the polished, intuitive UX of Termius combined with the access management depth of Teleport, the zero-trust architecture of HashiCorp Boundary, and the AI-augmented developer experience that none of the incumbents currently provide, at a price point accessible to individual developers while scaling commercially to enterprise.

Pricing Tiers: - **Free (Individual):** Unlimited hosts, unlimited sessions, 5 vault items, no team features, community support. - **Pro (Individual):** Unlimited vault, AI autocomplete, session recording (30-day retention), priority support. \$9/month. - **Team (per-seat):** All Pro features + shared vaults, team host management, collaboration, HelixTrack integration. \$15/seat/month. - **Enterprise:** All Team features + SAML/SCIM, FIDO2 enforcement, unlimited recording retention, compliance dashboard, SLA, dedicated support. Custom pricing. - **Self-hosted:** All Enterprise features, deploy in your own Kubernetes cluster, full source access for licensed modules. Custom pricing.

2. Complete System Architecture

3. Complete Microservices Catalog

3.1 Service Standard Structure

Every microservice follows this internal package layout:

```
services/<name>/ |—
go.mod |— go.sum |—
cmd/ |   |— server/
|      |—
main.go          #
Entrypoint: wire
dependencies, start HTTP/
gRPC servers |—
internal/ |   |—
domain/          #
Domain entities, value
objects, aggregate roots
|   |—
repository/          # DB
access (implements domain
```

2. Complete System Architecture

Repository interfaces)
| └─
service/ #
Business logic
(application services)
| └─
handler/ #
HTTP handlers (Gin) or
gRPC server
implementations | └─
middleware/ #
Service-local middleware
| └─
events/ #
Kafka producer/consumer
implementations | └─
commands/ #
RabbitMQ command handlers
└─
pkg/
Exported library code
(if any) └─
migrations/
Numbered SQL migration
files (00001_init.up.sql,
etc.) └─
config/
Configuration structs
and defaults └─
Dockerfile

3.2 Service: API Gateway

Module: helixterm.io/services/gateway **Port:** :8080 (HTTP), :8443 (HTTPS — handled by Ingress) **Replicas:** 10 (minimum in prod, HPA max 50)

Responsibilities

- Single ingress point for all client traffic.

- JWT validation against Auth Service JWKS endpoint.
- Per-user/per-IP rate limiting (Redis-backed).
- Upstream routing to all microservices.
- Circuit breaking on all upstream calls.
- WebSocket upgrade proxy for terminal and collaboration endpoints.
- Request tracing (OpenTelemetry) and metrics (Prometheus).
- CORS enforcement.
- gzip/brotli response compression.

External API

The Gateway is the external API — it routes every endpoint listed in Section 7.

Events Produced

- `helix.gateway.requests` (analytics — high-volume, 7-day retention, 12 partitions)

Events Consumed

None. The Gateway is stateless — it does not consume Kafka events.

Dependencies

All microservices (routing targets), Redis (rate limiting + JWKS cache).

Scalability

Stateless. Horizontal scaling via HPA (CPU threshold 60%). Connection pool to each upstream: min=10, max=100.

Failure Modes

- **Upstream circuit open:** Return 503 with `Retry-After: 30`. Log to `helix.gateway.circuit_open` Kafka topic.
 - **JWT validation failure:** 401 response.
 - **Rate limit exceeded:** 429 with `X-RateLimit-Limit`, `X-RateLimit-Remaining`, `X-RateLimit-Reset` headers.
-

3.3 Service: Auth Service

Module: helixterm.io/services/auth **Port:** :8081 **Database:** helixterm_auth (PostgreSQL dedicated instance) **Cache:** auth: prefix in Redis cluster

Responsibilities

- User authentication: password, FIDO2/WebAuthn, TOTP, OAuth2 (OIDC/SAML).
- Token lifecycle: issue access tokens (JWT EdDSA/Ed25519, TTL=15min), refresh tokens (opaque, TTL=30d), device tokens (JWT EdDSA/Ed25519, TTL=24h).
- Session management: track active sessions per user/device.
- FIDO2/WebAuthn credential registration and assertion.
- TOTP registration and verification.
- OAuth2 authorization server: authorization code flow with PKCE.
- SAML 2.0 IdP-initiated and SP-initiated SSO.
- SCIM directory synchronization (inbound provisioning from Okta, Azure AD, Google Workspace).

Dependencies

- digital.vasic.auth — OAuth2 flows, token management
- digital.vasic.security — password hashing (Argon2id), token signing key storage
- digital.vasic.cache — JWKS caching, session token cache
- digital.vasic.observability — traces and metrics
- digital.vasic.ratelimiter — login attempt rate limiting
- PostgreSQL (helixterm_auth)
- Redis

Key Go Code

```
// Package: helixterm.io/services/auth
// File: internal/service/auth_service.go
```

```
package service
```

```
import (
    "context"
    "time"
```

```

"digital.vasic.auth/pkg/oauth2"
"digital.vasic.auth/pkg/jwt"
"digital.vasic.security/pkg/argon2"
"digital.vasic.security/pkg/storage"
"digital.vasic.ratelimiter/pkg/limiter"
"helixterm.io/services/auth/internal/domain"
"helixterm.io/services/auth/internal/repository"
"helixterm.io/services/auth/internal/events"
)

type AuthService struct {
    userRepo      repository.UserRepository
    sessionRepo   repository.SessionRepository
    tokenRepo     repository.TokenRepository
    webauthnRepo  repository.WebAuthnRepository
    passwordHasher argon2.Hasher
    tokenManager  jwt.Manager
    rateLimiter   limiter.RateLimiter
    eventProducer events.Producer
    signingKeyID  string
}

func (s *AuthService) Login(ctx context.Context, req
    domain.LoginRequest) (*domain.LoginResponse, error) {
    // Rate limit: 5 attempts per 15 minutes per IP
    if err := s.rateLimiter.Allow(ctx, "login:"+req.IPAddress, 5,
        15*time.Minute); err != nil {
        return nil, domain.ErrRateLimitExceeded
    }

    user, err := s.userRepo.FindByEmail(ctx, req.Email)
    if err != nil {
        return nil, domain.ErrInvalidCredentials // constant-time
        response
    }

    if !s.passwordHasher.Verify(req.Password, user.PasswordHash) {
        s.eventProducer.Produce(ctx, events.LoginFailed{UserID:
            user.ID, IP: req.IPAddress})
        return nil, domain.ErrInvalidCredentials
    }

    if user.MFAEnabled {
        challenge := s.generateMFACHallenge(ctx, user.ID)
        return &domain.LoginResponse{
            RequiresMFA: true,

```

```

        MFACChallenge: challenge,
    }, nil
}

return s.issueTokens(ctx, user, req.DeviceID)
}

func (s *AuthService) issueTokens(ctx context.Context, user
    *domain.User, deviceID string) (*domain.LoginResponse,
    error) {
    accessToken, err :=
        s.tokenManager.IssueAccessToken(jwt.Claims{
            Subject:    user.ID.String(),
            OrgID:      user.OrgID.String(),
            Email:      user.Email,
            Roles:      user.Roles,
            DeviceID:   deviceID,
            ExpiresAt:  time.Now().Add(15 * time.Minute),
            IssuedAt:   time.Now(),
            Issuer:      "https://auth.helixterm.io",
            Audience:   []string{"https://api.helixterm.io"},
            KeyID:      s.signingKeyID,
        })
    if err != nil {
        return nil, err
    }

    refreshToken := s.tokenManager.IssueRefreshToken()
    if err := s.tokenRepo.Store(ctx, refreshToken, user.ID,
        deviceID, 30*24*time.Hour); err != nil {
        return nil, err
    }

    s.eventProducer.Produce(ctx, events.LoginSucceeded{
        UserID:    user.ID,
        DeviceID:  deviceID,
        IP:        "",
        Timestamp: time.Now(),
    })

    return &domain.LoginResponse{
        AccessToken:  accessToken,
        RefreshToken: refreshToken,
        ExpiresIn:    900,
        TokenType:    "Bearer",
    }, nil
}

```

```
    }, nil
  }
```

REST API (key endpoints)

See Section 7 for full details. Key endpoints: - POST /api/v1/auth/login - POST /api/v1/auth/logout - POST /api/v1/auth/refresh - GET /api/v1/auth/.well-known/jwks.json - POST /api/v1/auth/webauthn/begin-registration - POST /api/v1/auth/webauthn/finish-registration - POST /api/v1/auth/webauthn/begin-login - POST /api/v1/auth/webauthn/finish-login - POST /api/v1/auth/totp/enable - POST /api/v1/auth/totp/verify - POST /api/v1/auth/saml/sso - GET /api/v1/auth/sessions - DELETE /api/v1/auth/sessions/:session_id

Kafka Events Produced

Topic	Event	Trigger
helix.auth.authenticated	LoginSucceeded	Successful login
helix.auth.authenticated	LoginFailed	Failed login attempt
helix.auth.authenticated	TokenRefreshed	Token refresh
helix.auth.authenticated	LoggedOut	Logout
helix.audit.events	AuthAuditEvent	All auth events

Kafka Events Consumed

- helix.users.deleted → purge all sessions and tokens for deleted user

3.4 Service: User Service

Module: helixterm.io/services/user **Port:** :8082 **Database:** helixterm_users (PostgreSQL)

Responsibilities

- User CRUD: create, read, update, delete, soft-delete.
- User profile management: avatar, display name, timezone, locale, preferences.
- User search (by email, name, org) with Postgres full-text search.
- User onboarding workflow state machine.

- SCIM user provisioning endpoint (consumed by Auth Service for inbound sync).
- User preferences storage (encrypted fields via `digital.vasic.security`).

Key REST Endpoints

- GET `/api/v1/users/me`
 - PUT `/api/v1/users/me`
 - GET `/api/v1/users/:user_id`
 - GET `/api/v1/users` (search, admin only)
 - DELETE `/api/v1/users/:user_id` (admin only)
 - PUT `/api/v1/users/me/avatar`
 - GET `/api/v1/users/me/preferences`
 - PUT `/api/v1/users/me/preferences`
-

3.5 Service: Vault Service

Module: `helixterm.io/services/vault` **Port:** `:8083` **Database:** `helixterm_vault` (PostgreSQL)

Responsibilities

- Zero-knowledge encrypted storage for: SSH private keys, passwords, API tokens, TLS certificates, secret notes.
- Client-side encryption enforced: vault items are AES-256-GCM encrypted before transmission.
- Server stores only ciphertext + encrypted metadata.
- Vault item types: `ssh_key`, `password`, `api_token`, `tls_cert`, `secret_note`, `totp_secret`.
- Vault sharing: items can be shared to teams or specific users, encrypted for each recipient's public key.
- Vault item versioning: full history of changes with the ability to restore previous versions.
- Integration with `digital.vasic.security` for key wrapping.

Key Domain Types

```
// Package: helixterm.io/services/vault
// File: internal/domain/vault_item.go
```

```
package domain
```

```

import (
    "time"
    "github.com/google/uuid"
)

type VaultItemType string

const (
    VaultItemTypeSSHKey      VaultItemType = "ssh_key"
    VaultItemTypePassword    VaultItemType = "password"
    VaultItemTypeAPIToken    VaultItemType = "api_token"
    VaultItemTypeTLSCert     VaultItemType = "tls_cert"
    VaultItemTypeSecretNote  VaultItemType = "secret_note"
    VaultItemTypeTOTPSecret  VaultItemType = "totp_secret"
)

// VaultItem represents a zero-knowledge encrypted vault entry.
// The server NEVER sees plaintext. EncryptedData is opaque
// ciphertext.
type VaultItem struct {
    ID          uuid.UUID      `json:"id"`
    UserID      uuid.UUID      `json:"user_id"`
    OrgID       *uuid.UUID     `json:"org_id,omitempty"`
    Type        VaultItemType `json:"type"`
    Name        string         `json:"name"`           //
                        Plaintext label (not secret)
    EncryptedData []byte       `json:"encrypted_data"` //
                        AES-256-GCM ciphertext
    Nonce        []byte        `json:"nonce"`          // 12-byte GCM nonce
    KeyID        string        `json:"key_id"`          // Which
                        key encrypted this
    Tags         []string      `json:"tags"`
    FolderID     *uuid.UUID     `json:"folder_id,omitempty"`
    CreatedAt    time.Time      `json:"created_at"`
    UpdatedAt    time.Time      `json:"updated_at"`
    Version      int           `json:"version"`
}

```

Key REST Endpoints

- GET /api/v1/vault/items
- POST /api/v1/vault/items
- GET /api/v1/vault/items/:item_id
- PUT /api/v1/vault/items/:item_id

- DELETE /api/v1/vault/items/:item_id
 - GET /api/v1/vault/items/:item_id/versions
 - POST /api/v1/vault/items/:item_id/restore/:version
 - POST /api/v1/vault/items/:item_id/share
 - GET /api/v1/vault/folders
 - POST /api/v1/vault/folders
 - PUT /api/v1/vault/folders/:folder_id
 - DELETE /api/v1/vault/folders/:folder_id
-

3.6 Service: Host Service

Module: helixterm.io/services/host **Port:** :8084 **Database:** helixterm_hosts (PostgreSQL)

Responsibilities

- Host CRUD: create, read, update, delete SSH hosts and groups.
- Host connectivity: label, hostname, port, auth method, vault item reference (for credentials).
- Host groups/folders: nested hierarchy with inheritance of settings.
- Host labels, tags, notes.
- Host health ping: background goroutine pings hosts via SSH TCP connect to track availability.
- Import/export: CSV, Termius JSON, OpenSSH config formats.
- Bastion/jump host chains: multi-hop SSH chains.
- Host templates: define reusable host configuration templates.

Key REST Endpoints

- GET /api/v1/hosts
- POST /api/v1/hosts
- GET /api/v1/hosts/:host_id
- PUT /api/v1/hosts/:host_id
- DELETE /api/v1/hosts/:host_id
- POST /api/v1/hosts/:host_id/connect (initiates SSH session)
- GET /api/v1/hosts/:host_id/ping
- GET /api/v1/host-groups
- POST /api/v1/host-groups
- PUT /api/v1/host-groups/:group_id
- DELETE /api/v1/host-groups/:group_id
- POST /api/v1/hosts/import

• GET /api/v1/hosts/export

3.7 Service: SSH Proxy Service

Module: helixterm.io/services/ssh-proxy **Port:** :8090 (WebSocket/HTTP) **Database:** helixterm_ssh_proxy (PostgreSQL — connection state only) **Critical Path:** This is the highest-latency-sensitive service in the platform.

Responsibilities

- Broker SSH connections between the client (WebSocket) and the target SSH daemon (TCP).
- Support for: password auth, public key auth, SSH certificate auth (via PKI service).
- Support for bastion/jump host chains (proxy jump).
- Multiplex multiple logical channels over a single SSH connection (per RFC 4254).
- Integrate with `digital.vasic.containers` for container-native sessions (no SSH daemon required).
- Forward all terminal I/O to the Terminal Session Service.
- Enforce session-level access controls (verify the requesting user has permission to connect to the host via the Org/RBAC model).
- Support for SSH agent forwarding (with explicit user approval per session).
- SSH known-hosts verification and TOFU (Trust On First Use) policy.
- Connection keepalive (TCP keepalive + SSH keepalive packets every 30s).

Container-Native Session (via `digital.vasic.containers`)

```
// Package: helixterm.io/services/ssh-proxy
// File: internal/session/container_session.go
```

```
package session
```

```
import (
    "context"
    "io"

    "digital.vasic.containers/pkg/runtime"
    "digital.vasic.containers/pkg/exec"
```

```

    "helixterm.io/services/ssh-proxy/internal/domain"
)

type ContainerSession struct {
    runtime runtime.ContainerRuntime
    execConn exec.ExecConnection
    sessionID string
}

// NewContainerSession creates a terminal session into a container
// without SSH.
// Uses digital.vasic.containers ContainerRuntime abstraction.
func NewContainerSession(
    ctx context.Context,
    rt runtime.ContainerRuntime,
    req domain.ContainerConnectRequest,
) (*ContainerSession, error) {
    execReq := exec.ExecRequest{
        ContainerID: req.ContainerID,
        Command:
            []string{req.Shell}, // e.g. ["/bin/bash"] or ["/bin/sh"]
        TTY:        true,
        Stdin:       true,
        Stdout:      true,
        Stderr:      true,
        Env:         req.Env,
        WorkingDir:  req.WorkingDir,
        User:        req.User,
    }

    conn, err := rt.Exec(ctx, execReq)
    if err != nil {
        return nil, fmt.Errorf("container exec: %w", err)
    }

    return &ContainerSession{
        runtime:  rt,
        execConn: conn,
        sessionID: req.SessionID,
    }, nil
}

func (s *ContainerSession) ReadFrom(r io.Reader) (int64, error) {
    return s.execConn.Write(r)
}

```

```

func (s *ContainerSession) WriteTo(w io.Writer) (int64, error) {
    return s.execConn.Read(w)
}

func (s *ContainerSession) Resize(cols, rows uint16) error {
    return s.execConn.Resize(cols, rows)
}

func (s *ContainerSession) Close() error {
    return s.execConn.Close()
}

```

RabbitMQ Commands Consumed

- `helix.cmd.ssh.connect` — initiate new SSH session
- `helix.cmd.ssh.disconnect` — terminate SSH session

Kafka Events Produced

- `helix.sessions.started`
- `helix.sessions.terminated`

3.8 Service: Terminal Session Service

Module: `helixterm.io/services/terminal` **Port:** `:8091` (WebSocket)

Database: `helixterm_terminal` (PostgreSQL)

Responsibilities

- WebSocket server for terminal I/O proxying.
- Terminal buffer management: scrollbar buffer stored in Redis (ring buffer, 10,000 lines).
- xterm.js protocol compatibility (PTY resize, cursor positioning, color codes).
- I/O multiplexing for collaborative sessions (fan-out to all session observers via Kafka).
- Command detection: parse command boundaries (PS1 prompt detection) and emit command events.
- Idle session timeout enforcement (configurable per org, default 30min).

- Integration with Recording Service: stream session output to `helix.recordings.segments` Kafka topic.
- Integration with AI Service: pipe command and output to AI for real-time analysis.

WebSocket Message Protocol

```
// Package: helixterm.io/services/terminal
// File: internal/protocol/message.go
```

```
package protocol
```

```
type MessageType string
```

```
const (
    MsgInput      MessageType = "input"      // Client → server:
        keyboard input
    MsgOutput     MessageType = "output"     // Server → client:
        terminal output
    MsgResize     MessageType = "resize"     // Client → server:
        terminal size change
    MsgPing       MessageType = "ping"       // Keepalive
    MsgPong       MessageType = "pong"       // Keepalive response
    MsgCommand    MessageType = "command"    // Server → client:
        detected command boundary
    MsgError      MessageType = "error"      // Server → client:
        error notification
    MsgClose      MessageType = "close"      // Either direction:
        close session
    MsgAISuggest  MessageType =
        "ai_suggest" // Server → client: AI suggestion
)
```

```
type Message struct {
    Type      MessageType `json:"type"`
    Payload   interface{} `json:"payload"`
    Seq       uint64      `json:"seq"` // Monotonically
        increasing sequence number
    TS        int64      `json:"ts"` // Unix microseconds
}
```

```
type InputPayload struct {
    Data []byte `json:"data"` // Raw bytes (may contain ANSI
        escape sequences)
}
```

```
type OutputPayload struct {
```

```

Data    []byte `json:"data"`
Session string `json:"session_id"`
}

type ResizePayload struct {
    Cols uint16 `json:"cols"`
    Rows uint16 `json:"rows"`
}

```

3.9 Service: SFTP Service

Module: helixterm.io/services/sftp **Port:** :8092 **Database:** helixterm_sftp (PostgreSQL)

Responsibilities

- SFTP session management: open, close, list, navigate.
- File operations: read, write, rename, delete, mkdir, symlink, stat.
- Transfer queue management: persist transfer jobs, resume interrupted transfers.
- Checksum verification: SHA-256 on transfer completion.
- Directory sync: bidirectional rsync-style sync with conflict detection.
- Large file support: chunked upload/download with resumable transfers.
- Per-transfer audit events.
- Integration with SSH Proxy for underlying SSH connection.

Key REST Endpoints

- POST /api/v1/sftp/sessions
- DELETE /api/v1/sftp/sessions/:session_id
- GET /api/v1/sftp/sessions/:session_id/list?path=
- POST /api/v1/sftp/sessions/:session_id/upload
- GET /api/v1/sftp/sessions/:session_id/download?path=
- DELETE /api/v1/sftp/sessions/:session_id/file?path=
- POST /api/v1/sftp/sessions/:session_id/mkdir
- POST /api/v1/sftp/sessions/:session_id/rename
- GET /api/v1/sftp/transfers
- DELETE /api/v1/sftp/transfers/:transfer_id
- POST /api/v1/sftp/transfers/:transfer_id/pause

- POST /api/v1/sftp/transfers/:transfer_id/resume
-

3.10 Service: Port Forwarding Service

Module: helixterm.io/services/port-forward **Port:** :8093 **Database:** helixterm_port_forward (PostgreSQL)

Responsibilities

- Persist port forwarding rule catalog.
- Lifecycle management: start, stop, restart, auto-reconnect forwarding rules.
- Rule types: local, remote, dynamic (SOCKS5), reverse.
- Status tracking: active connections, bytes transferred, latency.
- Auto-reconnect with exponential backoff (via `digital.vasic.recovery`).
- Metrics emission: bytes in/out, connection latency per rule.
- Conflict detection: prevent binding the same local port twice.

Key REST Endpoints

- GET /api/v1/port-forwards
 - POST /api/v1/port-forwards
 - GET /api/v1/port-forwards/:rule_id
 - PUT /api/v1/port-forwards/:rule_id
 - DELETE /api/v1/port-forwards/:rule_id
 - POST /api/v1/port-forwards/:rule_id/start
 - POST /api/v1/port-forwards/:rule_id/stop
 - GET /api/v1/port-forwards/:rule_id/metrics
-

3.11 Service: Snippet Service

Module: helixterm.io/services/snippet **Port:** :8086 **Database:** helixterm_snippets (PostgreSQL)

Responsibilities

- Snippet CRUD: create, edit, delete, tag, search command snippets.
- Snippet types: shell commands, scripts, SQL queries, YAML/JSON configs.

- Snippet folders and namespaces (per-user, per-team, per-org).
 - Full-text search via PostgreSQL `tsvector/tsquery`.
 - Snippet parameterization: `{{parameter_name}}` substitution at execution time.
 - Snippet execution history: track when/where each snippet was used.
 - Version control: git-like versioning with diff and revert.
 - Import/export: JSON, shell script formats.
-

3.12 Service: Keychain Service

Module: `helixterm.io/services/keychain` **Port:** `:8101` (gRPC only — no REST; internal use by Vault and PKI services)

Responsibilities

- Hardware-backed key storage integration via `digital.vasic.security`.
 - Wrapping and unwrapping of vault encryption keys.
 - Key rotation: generate new wrapping key, re-encrypt all vault items, invalidate old key.
 - Support for: macOS Secure Enclave, Android Keystore, Windows DPAPI, Linux kernel keyring.
 - HSM integration (PKCS#11) for enterprise deployments.
 - Key escrow for organization admins (encrypted with org admin's public key).
-

3.13 Service: Workspace Service

Module: `helixterm.io/services/workspace` **Port:** `:8085` **Database:** `helixterm_workspaces` (PostgreSQL)

Responsibilities

- Workspace CRUD: named collections of hosts, snippets, vault items, and settings.
- Workspace templates: pre-configured workspaces for common environments (production, staging, dev).
- Workspace sharing within teams/orgs.

- Workspace environment variables: per-workspace environment variables injected into terminal sessions.
 - Quick-launch: single-click to open all hosts in a workspace.
 - Workspace activity feed: recent sessions, changes, team activity.
-

3.14 Service: Collaboration Service

Module: helixterm.io/services/collab **Port:** :8099 **Database:** helixterm_collab (PostgreSQL) **Cache:** Redis pub/sub for real-time message routing

Responsibilities

- Session sharing: invite users to observe or co-pilot active terminal sessions.
- Role enforcement: observer (read-only), co-pilot (write with approval), owner (full control).
- Participant management: join/leave, kick, role changes.
- CRDT-based terminal buffer sync: ensure all observers have consistent view.
- Broadcast mode: stream to hundreds/thousands of observers for training/demo.
- Chat sidebar: text chat within a collaborative session.
- WebSocket fan-out: distribute terminal output to all participants.
- Emit participant events to Kafka for audit trail.

Key WebSocket Events

// Participant joined

```
{"type": "participant_joined", "payload": {"user_id": "...",  
  "role": "observer", "ts": 1234567890}}
```

// Participant role changed

```
{"type": "role_changed", "payload": {"user_id": "...",  
  "old_role": "observer", "new_role": "co_pilot"}}
```

// Chat message

```
{"type": "chat", "payload": {"user_id": "...", "message":  
  "Looking at line 42", "ts": 1234567890}}
```

// Session ended by owner

```
{"type": "session_ended", "payload": {"reason": "owner_closed",  
  "ts": 1234567890}}
```

3.15 Service: Notification Service

Module: `helixterm.io/services/notification` **Port:** `:8096` **Database:** `helixterm_notifications` (PostgreSQL)

Responsibilities

- Multi-channel notification delivery: in-app, email, push (FCM/APNs), Slack, Webhooks.
- Notification templates: Mustache-based templates stored in PostgreSQL.
- User notification preferences: per-channel, per-event-type opt-in/opt-out.
- Notification deduplication (Redis-based, 5-minute window).
- Delivery tracking: sent, delivered, read status per notification.
- Batch notifications: aggregate multiple events into a single digest email.
- RabbitMQ consumer for command delivery.

RabbitMQ Queues Consumed

- `helix.notif.email`
- `helix.notif.push`
- `helix.notif.slack`
- `helix.notif.webhook`

Kafka Events Consumed

- `helix.sessions.started` → notify team members if configured
 - `helix.security.anomalies` → immediate security alert
 - `helix.auth.authenticated` → login from new device/location
-

3.16 Service: Audit Service

Module: `helixterm.io/services/audit` **Port:** `:8094` **Database:** `helixterm_audit` (PostgreSQL — partitioned by org + month)

Responsibilities

- Append-only, Merkle-chained audit log.
- Consume audit events from Kafka `helix.audit.events`.

- Store events in partitioned PostgreSQL table.
- Provide query API: filter by user, org, event type, time range, resource ID.
- Export: JSON, CSV, PDF audit reports.
- Compliance dashboard: SOC 2, ISO 27001, FedRAMP control evidence generation.
- Real-time audit event streaming via WebSocket (for SIEM integration).
- Audit event integrity verification: verify the Merkle chain for any given time range.
- Retention policy enforcement: auto-delete events older than org-configured retention period (minimum 365 days for enterprise).

Merkle Chain Implementation

```
// Package: helixterm.io/services/audit
// File: internal/chain/merkle.go
```

```
package chain
```

```
import (
    "crypto/sha256"
    "encoding/hex"
    "encoding/json"
)
```

```
type AuditEvent struct {
    ID          string    `json:"id"`
    OrgID       string    `json:"org_id"`
    UserID      string    `json:"user_id"`
    EventType   string    `json:"event_type"`
    ResourceID  string    `json:"resource_id"`
    Data        interface{} `json:"data"`
    OccurredAt  int64     `json:"occurred_at"` // Unix
                nanoseconds
    PrevHash    string    `json:"prev_hash"` // SHA-256 of
                previous event JSON
    Hash        string    `json:"hash"`       // SHA-256 of this event JSON
                (excluding hash field)
}
```

```
func ComputeHash(event AuditEvent) (string, error) {
    // Compute hash without the Hash field itself
    type hashable struct {
```

```

    ID          string    `json:"id"`
    OrgID       string    `json:"org_id"`
    UserID      string    `json:"user_id"`
    EventType   string    `json:"event_type"`
    ResourceID  string    `json:"resource_id"`
    Data        interface{} `json:"data"`
    OccurredAt  int64     `json:"occurred_at"`
    PrevHash    string    `json:"prev_hash"`
}

h := hashable{
    ID: event.ID, OrgID: event.OrgID, UserID: event.UserID,
    EventType: event.EventType, ResourceID: event.ResourceID,
    Data: event.Data, OccurredAt: event.OccurredAt, PrevHash:
    event.PrevHash,
}

b, err := json.Marshal(h)
if err != nil {
    return "", err
}

sum := sha256.Sum256(b)
return hex.EncodeToString(sum[:]), nil
}

```

3.17 Service: Analytics Service

Module: helixterm.io/services/analytics **Port:** :8095 **Database:** helixterm_analytics (PostgreSQL — time-series optimized, partitioned by week)

Responsibilities

- Consume analytics events from Kafka.
- Aggregate: session counts, command frequency, transfer volumes, login patterns.
- Dashboard data: serve pre-aggregated metrics for the UI analytics dashboard.
- Usage reporting: per-user, per-team, per-org usage breakdowns.
- Feature adoption tracking: which features are used, by how many users.
- SLO tracking: terminal latency p50/p95/p99, session establishment time.

- Export: CSV, JSON, API integration with Grafana (via Prometheus-compatible endpoint).
-

3.18 Service: AI/Autocomplete Service

Module: `helixterm.io/services/ai` **Port:** `:8098` **Database:** `helixterm_ai` (PostgreSQL — user preferences, history) **Cache:** Redis — suggestion cache (TTL=300s)

Responsibilities

- Command autocomplete: given current command prefix + shell context, return ranked completions.
- Command explanation: explain terminal output on request.
- Anomaly detection: stream analysis of command sequences for suspicious patterns.
- Runbook generation: generate multi-step runbooks from natural language descriptions.
- Incident assist: surface relevant runbooks and history during incidents.
- Model management: manage local ONNX model lifecycle (load, update, version).
- Privacy enforcement: no sensitive credential data is included in inference context; vault references are redacted before passing to any model.
- Telemetry: emit suggestion-accepted/rejected events to `helix.ai.suggestions` Kafka topic for model improvement.

Key REST Endpoints

- POST `/api/v1/ai/complete` — command completion
 - POST `/api/v1/ai/explain` — explain terminal output
 - POST `/api/v1/ai/runbook` — generate runbook
 - GET `/api/v1/ai/suggestions/history`
 - POST `/api/v1/ai/feedback` — accept/reject signal for a suggestion
-

3.19 Service: Session Recording Service

Module: `helixterm.io/services/recording` **Port:** `:8097` **Database:** `helixterm_recordings` (PostgreSQL — metadata only) **Storage:** S3-compatible object storage (MinIO in self-hosted, AWS S3 / GCS in cloud)

Responsibilities

- Consume recording segments from Kafka
`helix.recordings.segments`.
- Assemble segments into complete asciinema v2 format recordings.
- Store recordings in object storage with pre-signed URLs for playback.
- Sign recordings with Ed25519 private key for tamper detection.
- Provide playback API: stream recording as asciinema events.
- Search: full-text search over command output (indexing via PostgreSQL + tsvector).
- Retention enforcement: auto-delete recordings older than org-configured retention period.
- Export: download raw asciinema files, export as MP4 (server-side rendering via headless xterm.js).

Recording Signing

```
// Package: helixterm.io/services/recording
// File: internal/signing/signer.go
```

```
package signing
```

```
import (
    "crypto/ed25519"
    "crypto/sha256"
    "encoding/base64"
    "encoding/json"
)

type RecordingManifest struct {
    RecordingID string `json:"recording_id"`
    SessionID   string `json:"session_id"`
    UserID      string `json:"user_id"`
    StartedAt   int64  `json:"started_at"`
    EndedAt     int64  `json:"ended_at"`
    Checksum    string `json:"checksum"` // SHA-256 of recording
                  content
```

```

}

func Sign(manifest RecordingManifest, privKey ed25519.PrivateKey)
    (string, error) {
    data, err := json.Marshal(manifest)
    if err != nil {
        return "", err
    }
    sig := ed25519.Sign(privKey, data)
    return base64.StdEncoding.EncodeToString(sig), nil
}

func Verify(manifest RecordingManifest, sig string, pubKey
    ed25519.PublicKey) (bool, error) {
    data, err := json.Marshal(manifest)
    if err != nil {
        return false, err
    }
    sigBytes, err := base64.StdEncoding.DecodeString(sig)
    if err != nil {
        return false, err
    }
    return ed25519.Verify(pubKey, data, sigBytes), nil
}

```

3.20 Service: Certificate Authority Service (PKI)

Module: helixterm.io/services/pki **Port:** :8100 **Database:** helixterm_pki (PostgreSQL)

Responsibilities

- Issue short-lived SSH certificates signed by the organization's CA.
- CA key storage: CA private key stored in Vault Service (never in PKI service memory beyond signing operations).
- Certificate types: user certificates (for SSH login), host certificates (for server authentication).
- Certificate templates: define allowed principals, critical options, extensions per org/team/role.
- Certificate revocation: OCSP-like revocation list, checked by SSH Proxy on every connection.

- CA rotation: automated CA key rotation with overlap period.
- Certificate audit: every issued certificate is logged as an audit event.
- RabbitMQ command consumer for certificate issuance requests from SSH Proxy.

Certificate Issuance Flow

1. Client authenticates to SSH Proxy (JWT validated).
 2. SSH Proxy sends IssueCertificate RabbitMQ command.
 3. PKI Service retrieves CA signing key from Vault Service (over gRPC, mTLS).
 4. PKI signs the user's public key with the CA key.
 5. PKI returns the signed certificate to SSH Proxy.
 6. SSH Proxy presents the certificate to the target SSH server.
 7. Target SSH server verifies the certificate against the CA public key (pre-distributed via HelixTerminator's host provisioning agent).
-

3.21 Service: Organization/Team Service

Module: helixterm.io/services/org **Port:** :8087 **Database:** helixterm_org (PostgreSQL)

Responsibilities

- Multi-tenant organization management: Org → Teams → Members hierarchy.
- Org CRUD: create, update, delete organizations.
- Team management: create teams, manage membership, set team roles.
- RBAC model: define roles, assign roles to users within orgs/teams, define resource-level permissions.
- Permission evaluation: given a user + resource + action, evaluate whether the user is allowed.
- SCIM provisioning: expose SCIM v2 API for directory sync from Okta, Azure AD, etc.
- Invitation system: invite users to orgs/teams via email.
- Seat licensing: enforce per-org seat limits (via Billing Service).

RBAC Implementation

```
// Package: helixterm.io/services/org  
// File: internal/domain/rbac.go
```

```
package domain
```

```
type Permission string
```

```
const (
```

```
    // Host permissions
```

```
    PermHostView    Permission = "host:view"
```

```
    PermHostCreate  Permission = "host:create"
```

```
    PermHostEdit    Permission = "host:edit"
```

```
    PermHostDelete  Permission = "host:delete"
```

```
    PermHostConnect Permission = "host:connect"
```

```
    // Vault permissions
```

```
    PermVaultView    Permission = "vault:view"
```

```
    PermVaultCreate  Permission = "vault:create"
```

```
    PermVaultEdit    Permission = "vault:edit"
```

```
    PermVaultDelete  Permission = "vault:delete"
```

```
    PermVaultShare   Permission = "vault:share"
```

```
    // Session permissions
```

```
    PermSessionStart Permission = "session:start"
```

```
    PermSessionRecord Permission = "session:record"
```

```
    PermSessionShare  Permission = "session:share"
```

```
    PermSessionReplay Permission = "session:replay"
```

```
    // Audit permissions
```

```
    PermAuditView    Permission = "audit:view"
```

```
    PermAuditExport  Permission = "audit:export"
```

```
    // Admin permissions
```

```
    PermOrgAdmin     Permission = "org:admin"
```

```
    PermTeamAdmin    Permission = "team:admin"
```

```
    PermBillingView  Permission = "billing:view"
```

```
)
```

```
type Role struct {
```

```
    ID          string    `json:"id"`
```

```
    Name        string    `json:"name"`
```

```
    Permissions []Permission `json:"permissions"`
```

```
    IsBuiltIn   bool      `json:"is_built_in" // System roles  
                can't be deleted
```

```
}
```

```

var BuiltInRoles = []Role{
  {ID: "owner", Name: "Owner", Permissions:
    []Permission{PermOrgAdmin, PermTeamAdmin,
    PermHostConnect, PermVaultShare, PermAuditExport},
    IsBuiltIn: true},
  {ID: "admin", Name: "Admin", Permissions:
    []Permission{PermHostCreate, PermHostEdit,
    PermHostDelete, PermVaultCreate, PermVaultEdit,
    PermVaultDelete, PermTeamAdmin, PermAuditView},
    IsBuiltIn: true},
  {ID: "member", Name: "Member", Permissions:
    []Permission{PermHostView, PermHostConnect,
    PermVaultView, PermSessionStart, PermSessionRecord},
    IsBuiltIn: true},
  {ID: "viewer", Name: "Viewer", Permissions:
    []Permission{PermHostView, PermSessionReplay,
    PermAuditView}, IsBuiltIn: true},
  {ID: "devops", Name: "DevOps", Permissions:
    []Permission{PermHostConnect, PermHostCreate,
    PermHostEdit, PermVaultView, PermSessionStart,
    PermSessionRecord, PermSessionShare}, IsBuiltIn: true},
}

```

3.22 Service: Billing Service

Module: helixterm.io/services/billing **Port:** :8104 **Database:** helixterm_billing (PostgreSQL)

Responsibilities

- Subscription management: free, pro, team, enterprise tiers.
 - Seat management: count active seats, enforce limits.
 - Stripe integration: customer creation, subscription creation, payment method management, webhook handling.
 - Invoice generation and retrieval.
 - Usage metering: track feature usage against plan limits.
 - Trial management: 14-day trial for pro/team plans.
 - Dunning: handle failed payments, grace periods, seat downgrade on non-payment.
-

3.23 Service: Configuration Service

Module: helixterm.io/services/config **Port:** :8102 **Database:** helixterm_config (PostgreSQL)

Responsibilities

- Centralized configuration for all services (feature flags, operational parameters).
 - Per-org configuration overrides: e.g., session idle timeout, recording retention, MFA enforcement level.
 - Runtime config updates: services poll or subscribe to config updates via Kafka.
 - Config schema validation: prevent invalid configurations.
 - Config audit trail: every config change is an audit event.
-

3.24 Service: Health/Monitoring Service

Module: `helixterm.io/services/health` **Port:** `:8103`

Responsibilities

- Aggregate health status from all services (via `/health/live` and `/health/ready` endpoints).
 - Expose unified health dashboard API.
 - Alert routing: emit health alerts to Notification Service.
 - SLO tracking: calculate error budgets for each service.
 - Integration with Prometheus Alertmanager for paging.
-

3.25 Service: Container Registry Bridge

Module: `helixterm.io/services/container-bridge` **Port:** `:8105`

Database: `helixterm_container_bridge` (PostgreSQL)

Responsibilities

- Kubernetes cluster registration: manage kubeconfig references stored in Vault.
- Pod listing: list pods across registered clusters.
- Container shell sessions: initiate exec sessions into pods (via `digital.vasic.containers`).
- Container health dashboard: pod status, resource usage, events.
- Container log streaming: tail container logs in the terminal.
- Health monitoring: consume `helix.containers.health` Kafka events.

- Namespace management: list and navigate Kubernetes namespaces.
 - Docker/Podman host registration and local container management.
-

3.26 Service: HelixTrack Integration Service

Module: helixterm.io/services/helixtrack-bridge **Port:** :8106

Database: helixterm_helixtrack_bridge (PostgreSQL)

Responsibilities

- OAuth2 integration with helixtrack.ru/core using digital.vasic.auth.
- Link terminal sessions to HelixTrack issues and sprints.
- Import host configurations from HelixTrack environments.
- Sync deployment events from HelixTerminator sessions to HelixTrack timeline.
- Surface relevant HelixTrack issues in the terminal sidebar (context-aware: based on hostname, recent commands).
- Webhook receiver: receive HelixTrack issue state changes and surface them as notifications.

4. Database Architecture

5. Kafka & RabbitMQ Architecture

5.1 Messaging Technology Selection Rationale

Criterion	Apache Kafka	RabbitMQ
Delivery model	Log-based, replayable	Queue-based, consumed-once
Consumer groups	Multiple independent consumers	Competing consumers (load balanced)
Ordering	Guaranteed within partition	Per-queue FIFO
Retention	Configurable (days/weeks)	Until acknowledged
Use in HelixTerminator	Events (facts that happened)	Commands (actions to execute)
Throughput	Millions/sec	Hundreds of thousands/sec

Version: Apache Kafka 3.9 (per CANONICAL_FACTS.md CD-4; KRaft mode, no ZooKeeper), RabbitMQ 3.13.3.

5.2 Kafka Cluster Configuration

4. Database Architecture

```
yaml # deploy/kubernetes/kafka-cluster.yaml (using
Strimzi operator) apiVersion: kafka.strimzi.io/
v1beta2 kind: Kafka metadata: name: helix-kafka
namespace: helixterm-infra spec: kafka: version:
3.9.0 # per CANONICAL_FACTS.md CD-4 replicas:
9 # 3 per AZ, 3 AZs
listeners: - name: plain port: 9092 type: internal
tls: false - name: tls port: 9093 type: internal
tls: true authentication: type:
tls # mTLS for all internal
consumers config:
offsets.topic.replication.factor: 3
transaction.state.log.replication.factor: 3
transaction.state.log.min.isr: 2
default.replication.factor: 3 min.insync.replicas:
2 log.retention.hours: 168 # 7 days
default log.retention.bytes: -1 log.segment.bytes:
1073741824 # 1 GB segments compression.type:
snappy auto.create.topics.enable: false # Topics
created explicitly num.partitions: 12
num.recovery.threads.per.data.dir: 4 resources:
requests: memory: 8Gi cpu: 2 limits: memory: 16Gi
cpu: 4 storage: type: jbod volumes: - id: 0 type:
persistent-claim size: 1Ti class: fast-ssd
deleteClaim: false zookeeper: replicas:
0 # KRaft mode
```

5.3 Complete Kafka Topic Catalog

All topics are created via the digital.vasic.messaging topic management API, which calls the Kafka Admin API. Topic definitions are also committed as YAML to `deploy/kafka/topics/`.

```
yaml # deploy/kafka/topics/
helix.users.registered.yaml name:
helix.users.registered partitions: 6
replication_factor: 3 retention_ms: 604800000 #
7 days cleanup_policy: delete compression_type:
snappy min_insinc_replicas: 2
```

5.3.1 User and Auth Events

Topic	Partitions	Retention	Key	Purpose	
— — — — —	helix.users.registered	6	7d	user_id	New user registrations

4. Database Architecture

helix.users.updated | 6 | 7d | user_id | Profile updates | | helix.users.deleted | 6 | 30d | user_id | Account deletions (cascades) | | helix.auth.authenticated | 12 | 30d | user_id | All auth events (login, logout, refresh) | | helix.auth.mfa.events | 6 | 30d | user_id | MFA enable/disable/verify events | | helix.auth.tokens.revoked | 6 | 30d | token_id | Token revocation events |

5.3.2 Session Events

Topic	Partitions	Retention	Key	Purpose
helix.sessions.started	24	30d	session_id	SSH session start
helix.sessions.terminated	24	30d	session_id	SSH session end
helix.sessions.idle_timeout	12	7d	session_id	Session idle timeout events
helix.terminal.commands	48	30d	session_id	Individual command executions
helix.terminal.output	48	1d	session_id	Raw terminal output (high-volume, short retention)
helix.recordings.segments	48	7d	recording_id	Recording segment chunks
helix.recordings.completed	12	30d	recording_id	Recording finalization

5.3.3 Vault Events

Topic	Partitions	Retention	Key	Purpose
helix.vault.accessed	12	90d	user_id	Vault item access
helix.vault.created	12	90d	user_id	New vault items
helix.vault.updated	12	90d	item_id	Vault item changes
helix.vault.deleted	12	90d	item_id	Vault item deletions
helix.vault.shared	6	90d	item_id	Vault sharing events

5.3.4 Security Events

Topic	Partitions	Retention	Key	Purpose
helix.security.anomalies	12	90d	user_id	AI-detected anomalies
helix.security.alerts	12	90d	org_id	Security alerts
helix.pki.certificates.issued	6	365d	user_id	Certificate issuance

4. Database Architecture

helix.pki.certificates.revoked | 6 | 365d | serial |
Certificate revocation |

5.3.5 Audit Events

| Topic | Partitions | Retention | Key | Purpose | |
—|—|—|—|—| | helix.audit.events | 48 | 365d |
org_id | All audit events (compliance) |

5.3.6 Infrastructure Events

| Topic | Partitions | Retention | Key | Purpose | |
—|—|—|—|—| | helix.containers.health | 12 | 7d |
| cluster_id | Container/pod health changes | |
helix.portforward.opened | 12 | 7d | user_id | Port
forward opened | | helix.portforward.closed | 12 |
7d | user_id | Port forward closed | |
helix.sftp.transfers | 24 | 30d | user_id | SFTP
transfer events |

5.3.7 Analytics Events

| Topic | Partitions | Retention | Key | Purpose | |
—|—|—|—|—| | helix.analytics.events | 48 | 7d |
user_id | General analytics events | |
helix.gateway.requests | 48 | 7d | user_id | API
gateway request events | | helix.ai.suggestions | 12
| 30d | user_id | AI suggestion events |

5.3.8 Organization Events

| Topic | Partitions | Retention | Key | Purpose | |
—|—|—|—|—| | helix.orgs.created | 6 | 30d |
org_id | Org creation | | helix.orgs.updated | 6 | 30d
| org_id | Org updates | | helix.orgs.members.added |
12 | 30d | org_id | Member addition | |
helix.orgs.members.removed | 12 | 30d | org_id |
Member removal | | helix.teams.created | 6 | 30d |
org_id | Team creation | | helix.billing.events | 12
| 365d | org_id | All billing events |

5.4 Kafka Message Schemas

All Kafka messages use JSON encoding with an Avro-compatible schema registered in the Confluent Schema Registry. The Go struct definitions using `digital.vasic.messaging`:

```
``go // Package: helixterm.io/core // File: pkg/events/  
session_events.go
```

4. Database Architecture

```
package events
```

```
import "time"
```

```
// SessionStartedEvent is published to  
helix.sessions.started when an SSH session begins.  
type SessionStartedEvent struct { EventID string  
  json:"event_id" // UUID v7 EventType string  
  json:"event_type" // "session.started" SchemaVersion  
  int json:"schema_version" // 1 SessionID string  
  json:"session_id" UserID string json:"user_id" OrgID  
  string json:"org_id,omitempty" HostID string  
  json:"host_id" HostAddress string  
  json:"host_address" HostPort int json:"host_port"  
  Username string json:"username" AuthMethod string  
  json:"auth_method" ClientIP string json:"client_ip"  
  DeviceID string json:"device_id" CertSerial string  
  json:"cert_serial,omitempty" WorkspaceID string  
  json:"workspace_id,omitempty" OccurredAt time.Time  
  json:"occurred_at" }
```

```
// SessionTerminatedEvent is published to  
helix.sessions.terminated when an SSH session ends.  
type SessionTerminatedEvent struct { EventID string  
  json:"event_id" EventType string json:"event_type" //  
  "session.terminated" SchemaVersion int  
  json:"schema_version" // 1 SessionID string  
  json:"session_id" UserID string json:"user_id" OrgID  
  string json:"org_id,omitempty" HostID string  
  json:"host_id" Duration time.Duration  
  json:"duration_ms" CommandsCount int  
  json:"commands_count" BytesSent int64  
  json:"bytes_sent" BytesReceived int64  
  json:"bytes_received" ExitReason string  
  json:"exit_reason" // "user_close", "idle_timeout",  
  "error", "force_close" ErrorMessage string  
  json:"error_message,omitempty" RecordingID string  
  json:"recording_id,omitempty" OccurredAt time.Time  
  json:"occurred_at" }
```

```
// TerminalCommandEvent is published to  
helix.terminal.commands on each detected  
command. type TerminalCommandEvent struct  
{ EventID string json:"event_id" EventType string  
  json:"event_type" // "terminal.command" SessionID
```

4. Database Architecture

```
string json:"session_id" UserID string json:"user_id"
OrgID string json:"org_id,omitempty" HostID string
json:"host_id" Command string json:"command"
WorkingDir string json:"working_dir,omitempty"
ExitCode *int json:"exit_code,omitempty" DurationMs
int64 json:"duration_ms,omitempty" OccurredAt
time.Time json:"occurred_at" }
```

```
// AuditEvent is the canonical audit event published to
helix.audit.events. type AuditEvent struct { EventID
string json:"event_id" OrgID string json:"org_id"
UserID string json:"user_id,omitempty" ActorType
string json:"actor_type" // "user", "system", "service"
EventType string json:"event_type" ResourceType
string json:"resource_type,omitempty" ResourceID
string json:"resource_id,omitempty" ResourceName
string json:"resource_name,omitempty" Action string
json:"action" Outcome string json:"outcome" //
"success", "failure" IPAddress string
json:"ip_address,omitempty" UserAgent string
json:"user_agent,omitempty" Data interface{}
json:"data,omitempty" PrevHash string
json:"prev_hash" Hash string json:"hash" OccurredAt
time.Time json:"occurred_at" }
```

```
// SecurityAnomalyEvent is published to
helix.security.anomalies by the AI service. type
SecurityAnomalyEvent struct { EventID string
json:"event_id" SessionID string json:"session_id"
UserID string json:"user_id" OrgID string
json:"org_id" AnomalyType string
json:"anomaly_type" // "bulk_delete", "escalation",
"exfiltration", "unusual_time" Severity string
json:"severity" // "low", "medium", "high", "critical"
Command string json:"command,omitempty" Score
float64 json:"score" // 0.0-1.0 anomaly confidence
Explanation string json:"explanation" AutoAction
string json:"auto_action,omitempty" // "alert",
"suspend_session", "block" OccurredAt time.Time
json:"occurred_at" } ``
```

5.5 Kafka Consumer Groups

```
``go // Package: helixterm.io/services/audit // File:
internal/events/consumer.go
```

4. Database Architecture

package events

```
import ( "context" "digital.vasic.messaging/pkg/kafka"  
"helixterm.io/services/audit/internal/service" )
```

```
const ( ConsumerGroupAudit = "helix-audit-service"  
ConsumerGroupAnalytics = "helix-analytics-service"  
ConsumerGroupNotif = "helix-notification-service"  
ConsumerGroupAI = "helix-ai-service"  
ConsumerGroupRecording = "helix-recording-  
service" )
```

```
func StartAuditConsumer(ctx context.Context, svc  
service.AuditService) { consumer :=  
kafka.NewConsumer(kafka.ConsumerConfig{ Brokers:  
os.Getenv("KAFKA_BROKERS"), GroupID:  
ConsumerGroupAudit, Topics:  
[]string{"helix.audit.events"}, AutoCommit: false, //  
Manual offset commit after processing StartOffset:  
kafka.OffsetNewest, RetryAttempts: 3, RetryBackoff:  
kafka.ExponentialBackoff(100*time.Millisecond,  
5*time.Second), })
```

```
consumer.RegisterHandler("helix.audit.events",  
func(msg kafka.Message) error { var event  
AuditEvent if err := json.Unmarshal(msg.Value,  
&event); err != nil { return kafka.ErrUnmarshal //  
routes to DLQ } return svc.ProcessEvent(ctx, event) })  
consumer.Start(ctx) } ``
```

Consumer Group Assignments

Consumer Group	Topics	Services
helix-audit-service	helix.audit.events	Audit Service
helix-analytics-service	helix.analytics.events, helix.gateway.requests, helix.sessions.*, helix.terminal.commands	Analytics Service
helix-notification-service	helix.sessions.started, helix.security.anomalies, helix.auth.authenticated, helix.orgs.members.added	Notification Service
helix-ai-service	helix.terminal.commands, helix.terminal.output	AI Service
helix-recording-service	helix.recordings.segments	Recording Service
helix-user-cleanup	helix.users.deleted	Auth, Vault, Host, Org Services
helix-pki-audit	helix.pki.certificates.issued,	

4. Database Architecture

helix.pki.certificates.revoked | Audit Service (PKI events) | | helix-helixtrack-bridge |
helix.sessions.started, helix.sessions.terminated | HelixTrack Bridge |

5.6 Dead Letter Queue (DLQ) Handling

```
``go // Package: digital.vasic.messaging/pkg/kafka //  
Dead letter queue routing pattern used by all  
consumers  
  
type DLQHandler struct { producer kafka.Producer  
dlqSuffix string // “.dlq” by default }  
  
func (h *DLQHandler) Handle(msg kafka.Message, err  
error, attempt int) { dlqTopic := msg.Topic +  
h.dlqSuffix dlqMsg := kafka.Message{ Topic: dlqTopic,  
Key: msg.Key, Value: msg.Value, Headers:  
append(msg.Headers, kafka.Header{Key: “dlq-  
original-topic”, Value: []byte(msg.Topic)},  
kafka.Header{Key: “dlq-error”, Value:  
[]byte(err.Error())}, kafka.Header{Key: “dlq-attempt”,  
Value: []byte(strconv.Itoa(attempt))},  
kafka.Header{Key: “dlq-timestamp”, Value:  
[]byte(time.Now().Format(time.RFC3339Nano))}, ), }  
h.producer.Produce(dlqMsg) } ``
```

DLQ topics (auto-created with 30-day retention): -

helix.audit.events.dlq - helix.sessions.started.dlq
- helix.recordings.segments.dlq - etc.

DLQ messages are processed by a dedicated helix-dlq-reprocessor service that supports manual replay via admin API.

5.7 RabbitMQ Configuration

```
yaml # deploy/kubernetes/rabbitmq-cluster.yaml  
(using RabbitMQ Operator) apiVersion:  
rabbitmq.com/v1beta1 kind: RabbitmqCluster  
metadata: name: helix-rabbitmq namespace:  
helixterm-infra spec: replicas: 3 image:  
rabbitmq:3.13.3-management resources: requests:  
memory: 2Gi cpu: 1 limits: memory: 4Gi cpu: 2  
rabbitmq: additionalConfig: |  
vm_memory_high_watermark.relative = 0.7  
disk_free_limit.absolute = 5GB heartbeat = 30  
consumer_timeout = 300000 default_vhost =
```

4. Database Architecture

```
helixterm loopback_users.guest = false tls:  
secretName: rabbitmq-tls-secret caSecretName:  
rabbitmq-ca-secret
```

5.8 Complete RabbitMQ Exchange and Queue Catalog

Exchanges

```
helix.commands          (direct, durable) – Work  
queue commands helix.notifications    (direct,  
durable) – Notification delivery  
helix.events            (fanout, durable) –  
Internal event fanout helix.dlx  
(direct, durable) – Dead letter exchange
```

Queue Definitions (with arguments)

```
``go // Package: digital.vasic.messaging/pkg/  
rabbitmq // Declares all queues at service startup  
  
var queues = []QueueDeclaration{ // SSH commands  
{ Name: "helix.cmd.ssh.connect", Exchange:  
"helix.commands", RoutingKey: "ssh.connect",  
Durable: true, Args: amqp.Table{ "x-dead-letter-  
exchange": "helix.dlx", "x-dead-letter-routing-key":  
"dlq.ssh.connect", "x-message-ttl": int32(30000), // 30s  
— SSH connect must happen quickly "x-max-length":  
int64(10000), }, }, { Name: "helix.cmd.ssh.disconnect",  
Exchange: "helix.commands", RoutingKey:  
"ssh.disconnect", Durable: true, Args: amqp.Table{ "x-  
dead-letter-exchange": "helix.dlx", "x-dead-letter-  
routing-key": "dlq.ssh.disconnect", "x-message-ttl":  
int32(10000), "x-max-length": int64(10000), }, }, // SFTP  
commands { Name: "helix.cmd.sftp.transfer",  
Exchange: "helix.commands", RoutingKey:  
"sftp.transfer", Durable: true, Args: amqp.Table{ "x-  
dead-letter-exchange": "helix.dlx", "x-dead-letter-  
routing-key": "dlq.sftp.transfer", "x-message-ttl":  
int32(3600000), // 1 hour "x-max-length":  
int64(100000), "x-max-priority": int32(10), }, }, //  
Notification queues {Name: "helix.notif.email",  
Exchange: "helix.notifications", RoutingKey: "email",  
Durable: true, Args: amqp.Table{"x-message-ttl":  
int32(3600000), "x-dead-letter-exchange":  
"helix.dlx"}}, {Name: "helix.notif.push", Exchange:  
"helix.notifications", RoutingKey: "push", Durable:
```

4. Database Architecture

```
true, Args: amqp.Table{"x-message-ttl": int32(60000),
"x-dead-letter-exchange": "helix.dlx"}}, {Name:
"helix.notif.slack", Exchange: "helix.notifications",
RoutingKey: "slack", Durable: true, Args:
amqp.Table{"x-message-ttl": int32(300000), "x-dead-
letter-exchange": "helix.dlx"}}, {Name:
"helix.notif.webhook", Exchange: "helix.notifications",
RoutingKey: "webhook", Durable: true, Args:
amqp.Table{"x-message-ttl": int32(300000), "x-dead-
letter-exchange": "helix.dlx"}}, // PKI commands
{Name: "helix.cmd.pki.issue", Exchange:
"helix.commands", RoutingKey: "pki.issue", Durable:
true, Args: amqp.Table{"x-message-ttl":
int32(30000)}}, // Recording commands {Name:
"helix.cmd.recording.generate", Exchange:
"helix.commands", RoutingKey: "recording.generate",
Durable: true, Args: amqp.Table{"x-message-ttl":
int32(3600000)}}, // Container commands {Name:
"helix.cmd.container.exec", Exchange:
"helix.commands", RoutingKey: "container.exec",
Durable: true, Args: amqp.Table{"x-message-ttl":
int32(30000)}}, // HelixTrack commands {Name:
"helix.cmd.helixtrack.sync", Exchange:
"helix.commands", RoutingKey: "helixtrack.sync",
Durable: true, Args: amqp.Table{"x-message-ttl":
int32(300000)}}, // DLQ queues {Name:
"helix.dlq.ssh.connect", Exchange: "helix.dlx",
RoutingKey: "dlq.ssh.connect", Durable: true}, {Name:
"helix.dlq.sftp.transfer", Exchange: "helix.dlx",
RoutingKey: "dlq.sftp.transfer", Durable: true},
{Name: "helix.dlq.notifications", Exchange:
"helix.dlx", RoutingKey: "dlq.notifications", Durable:
true}, } ``
```

RabbitMQ Message Schemas

```
`go // SSH Connect Command type SSHConnectCommand
struct { CommandID
stringjson:"command_id"CommandType
stringjson:"command_type"// "ssh.connect"
SessionID    stringjson:"session_id"UserID
stringjson:"user_id"HostID
stringjson:"host_id"HostAddress
stringjson:"host_address"HostPort
```

4. Database Architecture

```
intjson:"host_port"Username
stringjson:"username"AuthMethod
stringjson:"auth_method"VaultItemRef
stringjson:"vault_item_ref,omitempty"// Vault item
ID (not the credential itself) CertRequest
booljson:"cert_request"// Request PKI-signed cert
PTYCols      intjson:"pty_cols"PTYRows
intjson:"pty_rows"Shell
stringjson:"shell"Timeout
intjson:"timeout_seconds"IssuedAt
time.Timejson:"issued_at"` }

// SFTP Transfer Command type
SFTPTransferCommand struct { CommandID string
json:"command_id" CommandType string
json:"command_type" // "sftp.transfer" TransferID
string json:"transfer_id" SessionID string
json:"transfer_id" UserID string json:"user_id"
Direction string json:"direction" // "upload" or
"download" LocalPath string json:"local_path"
RemotePath string json:"remote_path" ChunkSize
int64 json:"chunk_size" Checksum bool
json:"checksum" Priority int json:"priority" IssuedAt
time.Time json:"issued_at" }

// Notification Command type NotificationCommand
struct { CommandID string json:"command_id"
CommandType string json:"command_type" //
"notification.send" NotifID string json:"notif_id"
Channel string json:"channel" // "email", "push",
"slack", "webhook" UserID string json:"user_id"
TemplateID string json:"template_id" Data
interface{} json:"data" Priority string
json:"priority" // "normal", "high", "urgent" IssuedAt
time.Time json:"issued_at" } ``
```

6. Zero Trust Security Architecture

6.1 Zero Trust Principles

HelixTerminator's security model is built on the NIST Zero Trust Architecture (SP 800-207) guidelines:

1. **Never trust, always verify.** No implicit trust based on network location. Every request — whether from a client, an internal service, or an automated agent — must be authenticated and authorized before access is granted.
2. **Least privilege.** Every subject (user, service, agent) has only the permissions it needs for its current task, for the minimum duration necessary.
3. **Assume breach.** The system is designed to contain and detect breaches. A compromised service should not provide unrestricted lateral movement.
4. **Continuous verification.** Authentication is not a one-time event. Short token TTLs, certificate rotation, and behavioral analysis continuously re-validate trust.

6.2 SPIFFE/SPIRE Workload Identity

Architecture

SPIRE (SPIFFE Runtime Environment) is deployed as: - **SPIRE Server:** StatefulSet (3 replicas), PostgreSQL backend for registration entries. - **SPIRE Agent:** DaemonSet on every node. Attestation via Kubernetes Service Account tokens.

Every service pod has an Envoy sidecar injected by Istio that communicates with the SPIRE Agent via Unix domain socket to obtain X.509 SVIDs.

```
# SPIRE registration entry for auth-service
apiVersion: spire.spiffe.io/v1alpha1
kind: ClusterSPIFFEID
metadata:
  name: auth-service-id
spec:
```

```

selector:
  podLabel:
    app: auth-service
  spiffeIDTemplate: "spiffe://helixterm.io/ns/
    {{ .PodMeta.Namespace }}/sa/
    {{ .PodSpec.ServiceAccountName }}"
  ttl: 1h
  admin: false

```

SVIDs are rotated 5 minutes before expiry. If rotation fails, Envoy continues using the current SVID until expiry (5-minute grace), then the pod is restarted.

mTLS Enforcement

All inter-service communication uses mTLS with X.509 SVIDs. PeerAuthentication in STRICT mode ensures Envoy rejects any connection without a valid SPIFFE certificate.

```

apiVersion: security.istio.io/v1beta1
kind: PeerAuthentication
metadata:
  name: default
  namespace: helixterm-prod
spec:
  mtls:
    mode: STRICT

```

6.3 JWT Architecture

6.3.1 Access Token

```

// JWT Header
{
  "alg": "EdDSA",
  "typ": "JWT",
  "kid": "helix-2026-06-key-01" // key ID for JWKS lookup
}

// JWT Payload
{
  "iss": "https://auth.helixterm.io",
  "sub": "usr_01JXXXXXXXXXXXXXXXXXXXXXXX", // ULID
  "aud": ["https://api.helixterm.io"],

```

```

    "iat": 1751118000,
    "exp": 1751118900, // iat + 900s (15 minutes)
    "jti": "01JXXXXXXXXXXXXXXXXXXXXXXX", // JWT ID – nonce
        for replay prevention
    "email": "user@example.com",
    "org_id": "org_01JXXXXXXXXXXXXXXXXXXXXXXX",
    "roles": ["member", "devops"],
    "device_id": "dev_01JXXXXXXXXXXXXXXXXXXXXXXX",
    "session_id": "ses_01JXXXXXXXXXXXXXXXXXXXXXXX",
    "scope": "read write admin"
}

```

Access tokens are **EdDSA-signed** (Ed25519 key pair), per CANONICAL_FACTS.md CD-7. The public key is exposed at GET /api/v1/auth/.well-known/jwks.json. The Gateway validates tokens locally using cached JWKS (TTL=300s in Redis). Ed25519 signing keys are rotated every 90 days with a 7-day overlap period for graceful token expiry.

6.3.2 Refresh Token

Refresh tokens are opaque random 32-byte values, stored as SHA-256 hash in PostgreSQL. They are:

- Single-use (rotation on every use).
- Replaced with a new token on each use (refresh token rotation).
- Invalidated if a previously-used token is presented (token reuse detection → all sessions for the user revoked).
- TTL: 30 days from issuance.

6.3.3 Device Token

Device tokens are JWT EdDSA/Ed25519 tokens with TTL=24h, issued during device registration. They carry device_cert claim — the fingerprint of the device's X.509 certificate issued by the PKI service. Device tokens can be used for automation/CI pipelines where interactive MFA is not possible.

6.3.4 JWT Key Rotation

```

// Package: helixterm.io/services/auth
// File: internal/keys/rotation.go

```

```
package keys
```

```
import (
```

```

"crypto/ed25519"
"crypto/rand"
"encoding/json"
"time"

"digital.vasic.security/pkg/storage"
"gopkg.in/square/go-jose.v2"
)

type KeyRotationManager struct {
    store      storage.SecureStore
    activeKeyID string
    keys       map[string]ed25519.PrivateKey
    mu         sync.RWMutex
}

func (m *KeyRotationManager) Rotate() error {
    // EdDSA (Ed25519) signing key, per CANONICAL_FACTS.md CD-7
    _, newKey, err := ed25519.GenerateKey(rand.Reader)
    if err != nil {
        return err
    }
    newKeyID := fmt.Sprintf("helix-%s-key-%s",
        time.Now().Format("2006-01"),
        time.Now().Format("02"))

    // Store encrypted in secure storage via
    // digital.vasic.security
    if err := m.store.Store(newKeyID, newKey); err != nil {
        return err
    }

    m.mu.Lock()
    m.keys[newKeyID] = newKey
    m.activeKeyID = newKeyID
    // Keep previous key for 7 days (grace period)
    m.mu.Unlock()

    return nil
}

func (m *KeyRotationManager) JWKS() jose.JSONWebKeySet {
    m.mu.RLock()
    defer m.mu.RUnlock()
    var keys []jose.JSONWebKey

```

```

    for id, key := range m.keys {
        keys = append(keys, jose.JSONWebKey{
            Key:      key.Public(),
            KeyID:     id,
            Algorithm: "EdDSA",
            Use:      "sig",
        })
    }
    return jose.JSONWebKeySet{Keys: keys}
}

```

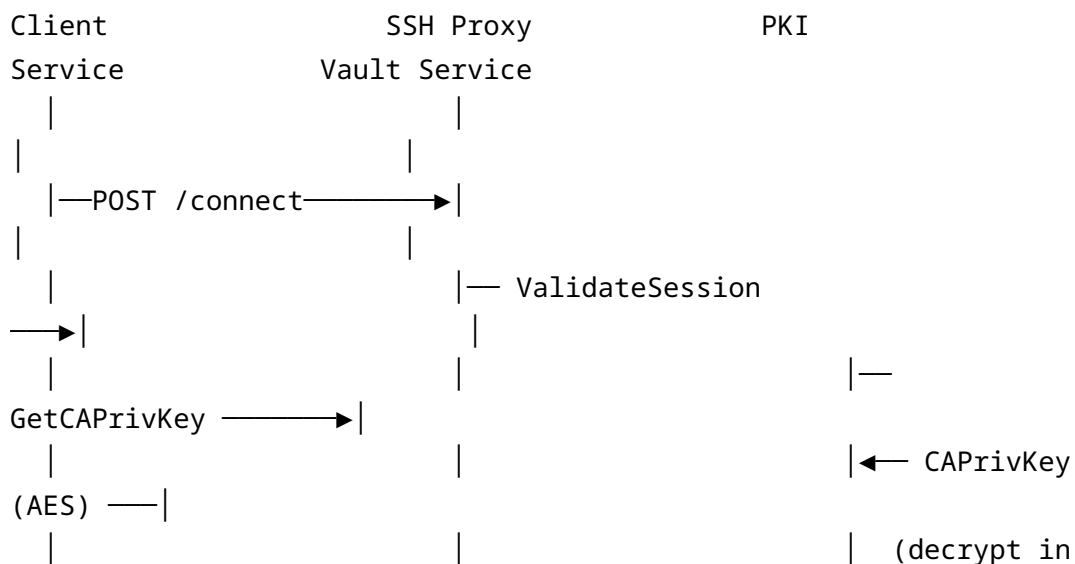
6.4 SSH Certificate Authority Architecture

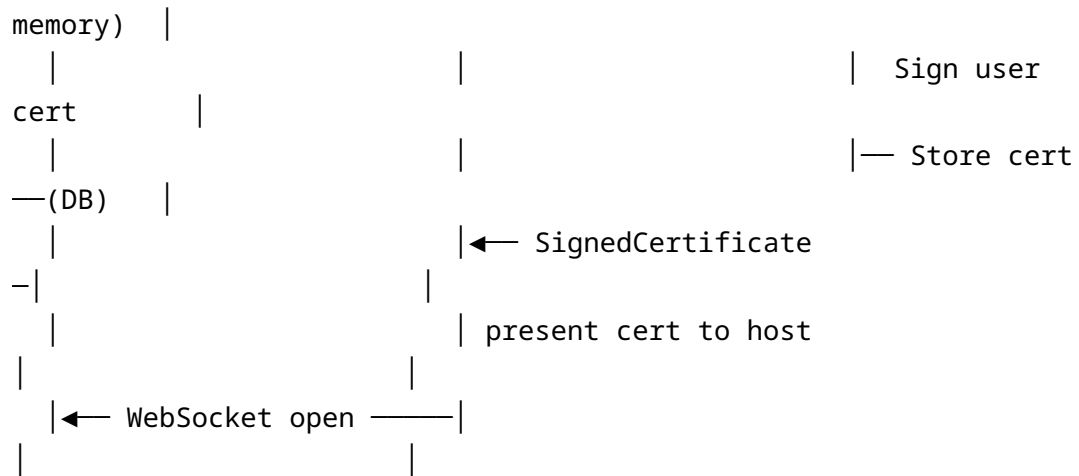
Why SSH Certificates Instead of Static Keys

Static SSH private keys are long-lived, hard to revoke, and accumulate on target hosts in `authorized_keys` files. HelixTerminator's PKI service eliminates this by:

1. Distributing only the **CA public key** to target hosts (via `TrustedUserCAKeys` in `/etc/ssh/sshd_config`).
2. Issuing **short-lived user certificates** (TTL=8h default, configurable 15min–24h) signed by the CA.
3. Storing the **CA private key** in the Vault service (zero-knowledge encrypted), retrieving it only for signing operations.
4. **Revoking** certificates via a KRL (Key Revocation List) distributed to target hosts.

Certificate Issuance Flow (detailed)





SSH Certificate Configuration

```
// Package: helixterm.io/services/pki
// File: internal/ca/signer.go
```

```
package ca
```

```
import (
    "crypto/ed25519"
    "crypto/rand"
    "time"

    "golang.org/x/crypto/ssh"
)
```

```
type CertificateSigner struct {
    caSigner  ssh.Signer
    keyID     string
}
```

```
func (s *CertificateSigner) IssueUserCert(
    userPublicKey ssh.PublicKey,
    userEmail     string,
    principals    []string,
    ttl           time.Duration,
    extensions    map[string]string,
) (*ssh.Certificate, error) {
    cert := &ssh.Certificate{
        Key:          userPublicKey,
        CertType:     ssh.UserCert,
        KeyId:        userEmail,
        ValidPrincipals: principals,
```

```

ValidAfter:      uint64(time.Now().Unix()),
ValidBefore:     uint64(time.Now().Add(ttl).Unix()),
Permissions: ssh.Permissions{
    CriticalOptions: map[string]string{
        // "source-address": "0.0.0.0/0", // restrict by
IP if needed
    },
    Extensions: mergeExtensions(map[string]string{
        "permit-X11-forwarding": "",
        "permit-agent-forwarding": "",
        "permit-port-forwarding": "",
        "permit-pty": "",
        "permit-user-rc": "",
    }, extensions),
},
}
if err := cert.SignCert(rand.Reader, s.caSigner); err != nil {
    return nil, fmt.Errorf("sign cert: %w", err)
}
return cert, nil
}

```

6.5 FIDO2/WebAuthn

HelixTerminator implements FIDO2/WebAuthn using the go-webauthn/webauthn library (v0.10.2).

Supported Authenticators

- **Platform authenticators:** Touch ID (macOS), Face ID (iOS), Windows Hello, Android biometrics.
- **Roaming authenticators:** YubiKey 5 series (NFC/USB), Google Titan, any CTAP2-compliant FIDO2 key.
- **Discoverable credentials (passkeys):** Stored in platform credential manager (iCloud Keychain, Windows Credential Store, Android Credential Manager).

Registration Flow

```

// Package: helixterm.io/services/auth
// File: internal/handler/webauthn_handler.go

```

```
package handler
```

```

import (
    "github.com/go-webauthn/webauthn/webauthn"
    "github.com/gin-gonic/gin"
)

var WebAuthnConfig = &webauthn.Config{
    RPDisplayName: "HelixTerminator",
    RPID:         "helixterm.io",
    RPOrigins:     []string{"https://app.helixterm.io", "https://helixterm.io"},
    AttestationPreference: protocol.PreferDirectAttestation,
    AuthenticatorSelection: protocol.AuthenticatorSelection{
        AuthenticatorAttachment: protocol.CrossPlatform,
        ResidentKey:              protocol.ResidentKeyRequirementPreferred,
        UserVerification:         protocol.VerificationRequired,
    },
    Timeout: 60000, // 60 seconds
}

func (h *WebAuthnHandler) BeginRegistration(c *gin.Context) {
    user := c.MustGet("auth_user").(*domain.User)

    webauthnUser := &WebAuthnUser{
        ID:         []byte(user.ID.String()),
        Name:        user.Email,
        DisplayName: user.DisplayName,
    }

    options, session, err :=
        h.webauthn.BeginRegistration(webauthnUser,
            webauthn.WithResidentKeyRequirement(protocol.ResidentKeyRequirementPreferred),
            webauthn.WithAuthenticatorSelection(WebAuthnConfig.AuthenticatorSelection))

    if err != nil {
        c.JSON(500, gin.H{"error": err.Error()})
        return
    }

    // Store session in Redis (TTL=300s)
    h.cache.Set(c, "webauthn:reg:"+user.ID.String(), session,
        300*time.Second)

    c.JSON(200, options)
}

```

MFA Enforcement Policy

```
// Package: helixterm.io/services/auth
// File: internal/policy/mfa_policy.go

type MFAPolicy struct {
    OrgEnforcement string // "none", "optional", "required"
    ExemptRoles    []string
    ExemptDevices  []string // Trusted device IDs
    GracePeriodDays int      // Days after enforcement change
                    before users must enroll
}

func (p *MFAPolicy) RequiresMFA(user *domain.User, orgPolicy
    string) bool {
    if orgPolicy == "required" {
        for _, role := range p.ExemptRoles {
            if slices.Contains(user.Roles, role) {
                return false
            }
        }
        return true
    }
    return user.MFAEnabled // User's own preference
}
```

6.6 Vault Encryption Architecture (Zero Knowledge)

Client Side

Server Side

User password

|

▼

Argon2id(password, salt, t=3, m=65536, p=4)

| → 256-bit Master Key

|

▼

AES-256-GCM.Encrypt(vault_item_plaintext)

| → ciphertext + nonce

|

▼ HTTPS POST

nonce + key_id	Receive ciphertext +
(encrypted_data, nonce)	Store in PostgreSQL
or plaintext	NEVER has the Master Key

Key Derivation Specification

```
// Package: helixterm.io/core
// File: pkg/crypto/vault_kdf.go

package crypto

import (
    "crypto/rand"
    "crypto/sha256"

    "golang.org/x/crypto/argon2"
)

const (
    Argon2TimeCost    = 3
    Argon2MemoryCost = 65536 // 64 MiB
    Argon2Threads     = 4
    Argon2KeyLen      = 32    // 256-bit
    SaltLen           = 32    // 256-bit salt
)

type VaultKeyDerivation struct{}

func (VaultKeyDerivation) DeriveKey(password, salt []byte) []byte
{
    return argon2.IDKey(password, salt, Argon2TimeCost,
        Argon2MemoryCost, Argon2Threads, Argon2KeyLen)
}

func (VaultKeyDerivation) GenerateSalt() ([]byte, error) {
    salt := make([]byte, SaltLen)
    _, err := rand.Read(salt)
    return salt, err
}

// EncryptVaultItem encrypts a plaintext vault item for server
// storage.
```

```
// The encrypted_data field stored on the server is the result of
// this function.
func EncryptVaultItem(key, plaintext []byte) (ciphertext, nonce
    []byte, err error) {
    block, err := aes.NewCipher(key)
    if err != nil {
        return nil, nil, err
    }
    gcm, err := cipher.NewGCM(block)
    if err != nil {
        return nil, nil, err
    }
    nonce = make([]byte, gcm.NonceSize())
    if _, err = rand.Read(nonce); err != nil {
        return nil, nil, err
    }
    ciphertext = gcm.Seal(nil, nonce, plaintext, nil)
    return ciphertext, nonce, nil
}
```

Vault Key Wrapping for Sharing

When a vault item is shared with another user or team, the symmetric item key is encrypted with the recipient's public key (X25519 ECDH key exchange + AES-256-GCM):

```
// Package: helixterm.io/core
// File: pkg/crypto/vault_sharing.go

import "golang.org/x/crypto/nacl/box"

// WrapKeyForRecipient encrypts the item's symmetric key for a
// recipient.
// Uses NaCl box (Curve25519 ECDH + XSalsa20-Poly1305).
func WrapKeyForRecipient(
    senderPrivKey, recipientPubKey *[32]byte,
    itemKey []byte,
) ([]byte, error) {
    var nonce [24]byte
    if _, err := rand.Read(nonce[:]); err != nil {
        return nil, err
    }
    encrypted := box.Seal(nonce[:], itemKey, &nonce,
        recipientPubKey, senderPrivKey)
```

```

    return encrypted, nil
}

```

6.7 RBAC Implementation Details

The Org Service implements a four-level RBAC model:

Organization

- └ Role (built-in or custom)
 - └ Permissions (coarse-grained capabilities)
 - └ Resource ACLs (fine-grained per-resource)

Permission Evaluation

```

// Package: helixterm.io/services/org
// File: internal/service/permission_service.go

package service

type PermissionService struct {
    roleRepo      repository.RoleRepository
    memberRepo    repository.MemberRepository
    resourceACLRepo repository.ResourceACLRepository
    cache         cache.Cache
}

func (s *PermissionService) Authorize(ctx context.Context, req
    AuthorizeRequest) (bool, error) {
    cacheKey := fmt.Sprintf("perm:%s:%s:%s:%s", req.UserID,
        req.OrgID, req.Permission, req.ResourceID)

    // Check cache first (TTL=60s)
    if cached, err := s.cache.Get(ctx, cacheKey); err == nil {
        return cached == "1", nil
    }

    // 1. Get user's roles in this org
    member, err := s.memberRepo.FindByUserAndOrg(ctx, req.UserID,
        req.OrgID)
    if err != nil {
        return false, nil // Not a member → deny
    }

    // 2. Get role permissions

```

```

role, err := s.roleRepo.FindByID(ctx, member.RoleID)
if err != nil {
    return false, err
}

// 3. Check coarse-grained permission
hasPermission := slices.Contains(role.Permissions,
    string(req.Permission))

// 4. If coarse permission exists and resource ID is
    specified, check resource ACL
if hasPermission && req.ResourceID != "" {
    acl, err := s.resourceACLRepo.FindByResourceAndUser(ctx,
        req.ResourceID, req.UserID)
    if err != nil && req.ResourceType == "host" {
        // Host-specific ACL check
        hostPerm, err :=
            s.resourceACLRepo.FindHostPermission(ctx, req.ResourceID,
                req.UserID, req.OrgID)
        if err == nil {
            hasPermission =
                slices.Contains(hostPerm.Permissions,
                    string(req.Permission))
        }
    } else if err == nil {
        hasPermission = slices.Contains(acl.Permissions,
            string(req.Permission))
    }
}

// Cache the result
val := "0"
if hasPermission {
    val = "1"
}
s.cache.Set(ctx, cacheKey, val, 60*time.Second)

return hasPermission, nil
}

```

6.8 SOC 2 Type II Compliance Controls

Control Domain	Control	HelixTerminator Implementation
CC6.1		

Control Domain	Control	HelixTerminator Implementation
CC6.2	Logical access controls	RBAC + FIDO2 MFA enforcement
	Registration/deregistration	SCIM + org member lifecycle
CC6.3	Removal of access	Session revocation, token revocation on member removal
CC6.7	Restriction of system access	Zero-trust AuthorizationPolicy, mTLS
CC7.1	Detection of unauthorized access	AI anomaly detection, login alerts
CC7.2	Monitoring	Audit log + Prometheus + OpenTelemetry
CC7.3	Evaluation of security events	Audit Service + real-time alerts
CC8.1	Change management	GitOps, PR gates, HelixConstitution CI checks
A1.1	Availability	HPA, PDB, multi-AZ, circuit breakers
C1.1	Confidentiality	AES-256-GCM vault, mTLS, zero-knowledge architecture
PI1.1	Processing integrity	Merkle-chained audit log, checksum on all transfers

6.9 Network Security: Zero Trust Network Policies

```
# Default deny all ingress and egress
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
```

```

    name: default-deny-all
    namespace: helixterm-prod
spec:
  podSelector: {}
  policyTypes:
    - Ingress
    - Egress

---
# Auth service: allow ingress from gateway, egress to postgres and
# redis
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: auth-service-netpol
  namespace: helixterm-prod
spec:
  podSelector:
    matchLabels:
      app: auth-service
  policyTypes:
    - Ingress
    - Egress
  ingress:
    - from:
        - podSelector:
            matchLabels:
              app: gateway-service
      ports:
        - protocol: TCP
          port: 8081
  egress:
    - to:
        - namespaceSelector:
            matchLabels:
              kubernetes.io/metadata.name: helixterm-infra
      ports:
        - protocol: TCP
          port: 5432 # PostgreSQL
        - protocol: TCP
          port: 6379 # Redis
        - protocol: TCP
          port: 9093 # Kafka TLS
        - protocol: TCP

```

```

    port: 5671    # RabbitMQ TLS
  - to:
    - podSelector:
        matchLabels:
          app: spire-agent    # SPIFFE SVID socket

```

6.10 Biometric Authentication Flow

On Flutter mobile/desktop clients, the platform's biometric API is used as an authenticator:

```

// Flutter client: lib/auth/biometric_auth.dart

import 'package:local_auth/local_auth.dart';
import 'package:flutter_secure_storage/
  flutter_secure_storage.dart';

class BiometricAuthService {
  final LocalAuthentication _auth = LocalAuthentication();
  final FlutterSecureStorage _storage = const
    FlutterSecureStorage();

  Future<bool> authenticateWithBiometrics() async {
    final bool isAvailable = await _auth.canCheckBiometrics;
    if (!isAvailable) return false;

    final bool authenticated = await _auth.authenticate(
      localizedReason: 'Authenticate to access HelixTerminator',
      options: const AuthenticationOptions(
        stickyAuth: true,
        biometricOnly: false, // Allow PIN fallback
        sensitiveTransaction: true,
      ),
    );

    if (authenticated) {
      // Retrieve device token from secure storage, unlocked by
      // biometric
      final String? deviceToken = await _storage.read(
        key: 'helix_device_token',
        iOptions: const IOSOptions(accessibility:
          KeychainAccessibility.biometricCurrentSet),
        aOptions: const
          AndroidOptions(encryptedSharedPreferences: true),
      );
    }
  }
}

```

```

        return deviceToken != null;
    }
    return false;
}
}

```

The device token (JWT, TTL=24h) is stored in the platform's hardware-backed secure storage (Secure Enclave on iOS/macOS, Android Keystore, Windows DPAPI on Windows), accessible only after successful biometric authentication.

7. API Specification

8. Performance Architecture

8.1 Performance Targets

Metric	Target	Notes
API p50 latency	< 10ms	Auth, User, Host, Vault list endpoints
API p95 latency	< 50ms	All read endpoints
API p99 latency	< 200ms	Including cache misses
SSH session establishment	< 500ms	Including PKI cert issuance
Terminal keystroke → echo	< 30ms	End-to-end over WebSocket
WebSocket connections	10M+ concurrent	Across all terminal and collab sessions
API requests per second	5M+ rps	Across all services
Kafka throughput	500M events/day	Audit + terminal events combined
Session recording	100K+ concurrent	Recording segments to Kafka
SFTP transfer	1 Gbps+	Per-connection max

8.2 Horizontal Scaling Strategy Per Service

Every service is stateless (state in PostgreSQL/Redis) and scales horizontally via Kubernetes HPA.

Service	Min Replicas	Max Replicas	HPA Metric	Target
Gateway	10	100	CPU	60%
Auth	5	50	CPU + RPS	70% / 10K rps
User	3	30	CPU	70%
Vault	3	30	CPU	70%
Host	3	30	CPU	70%
SSH Proxy	20	500	Active connections	5000 conn/pod
Terminal	50	2000	Active WebSocket conn	5000 conn/pod
SFTP	5	100	CPU + conn	70% / 200 conn
Port Forward	5	50	Active rules	1000 rules/pod
Collab	10	200	Active sessions	2000 sessions/pod
Audit	5	50	Kafka lag	< 10K messages
Analytics	5	50	Kafka lag	< 10K messages
AI	10	100	CPU	50% (model inference heavy)
Recording	10	100	Kafka lag	< 5K segments
Notification	3	30	Queue depth	< 1K messages
PKI	3	20	RPS	< 5K rps
Org	3	30	CPU	70%
Billing	2	10	CPU	60%

8.3 Redis Caching Strategy (L1/L2)

Client → [L1: in-process memory cache (ristretto, 100MB)] → [L2: Redis Cluster] → PostgreSQL

L1 Cache (in-process, using `digital.vasic.cache`)

7. API Specification

```go // Package: digital.vasic.cache/pkg/l1 // In-process cache using ristretto // Used in: Auth (JWKS), Gateway (permission checks), Org (role permissions)

import "github.com/dgraph-io/ristretto"

var l1 = cache.NewL1Cache(cache.L1Config{ MaxCost: 100 \* 1024 \* 1024, // 100 MB NumCounters: 1\_000\_000, BufferItems: 64, })

// JWKS: L1 TTL=60s, L2 TTL=300s // Role permissions: L1 TTL=30s, L2 TTL=60s // Org config: L1 TTL=30s, L2 TTL=60s // Host list: L1 TTL=10s, L2 TTL=30s ```

### L2 Cache (Redis Cluster)

Redis Cluster with 6 nodes (3 primaries, 3 replicas), deployed in 3 AZs. Total memory: 96GB.

**Connection pool configuration:** ```go import "github.com/redis/go-redis/v9"

rdb := redis.NewClusterClient(&redis.ClusterOptions{ Addrs: strings.Split(os.Getenv("REDIS\_ADDRS"), ","), Password: os.Getenv("REDIS\_PASSWORD"), PoolSize: 100, // Per node MinIdleConns: 10, MaxRetries: 3, DialTimeout: 5 \* time.Second, ReadTimeout: 3 \* time.Second, WriteTimeout: 3 \* time.Second, TLSConfig: &tls.Config{...}, // mTLS }) ```

### Read Replica Strategy (PostgreSQL)

All SELECT queries that are not within a transaction use read replicas. Each service that uses `digital.vasic.database` automatically routes reads to the replica pool.

```go // Package: digital.vasic.database/pkg/db // Automatic read/write splitting  
type DB struct { primary *pgxpool.Pool* replicas []*pgxpool.Pool* idx atomic.Uint64 // Round-robin replica selection }

func (d *DB) QueryRow(ctx context.Context, sql string, args ...any) *pgx.Row* { replica := d.replicas[d.idx.Add(1)%uint64(len(d.replicas))] return replica.QueryRow(ctx, sql, args...) }

func (d *DB) Exec(ctx context.Context, sql string, args ...any) (*pgconn.CommandTag*, error) { return d.primary.Exec(ctx, sql, args...) } ```

PostgreSQL Connection Pool Configuration

```go // Per service primary connection pool primary := *pgxpool.New*(ctx, &*pgxpool.Config*{ ConnConfig: *pgxConfig*, MaxConns: 100, MinConns: 10, MaxConnLifetime: 30 \* time.Minute, MaxConnIdleTime: 10 \* time.Minute, HealthCheckPeriod: 30 \* time.Second, })

// Per service replica pool (round-robin across 2 replicas) replicaConfigs := []string{ os.Getenv("PG\_REPLICA\_1\_URL"), os.Getenv("PG\_REPLICA\_2\_URL"), } // Each replica: MaxConns=200 (heavier read load) ```

## 8.4 WebSocket Connection Management at Scale

---

## # 7. API Specification

---

The Terminal and Collab services need to handle millions of concurrent WebSocket connections. This is achieved via:

### ### 1. Gorilla WebSocket Upgrade + Custom Multiplexer

Each Terminal pod handles up to 5,000 concurrent WebSocket connections via Go's efficient goroutine scheduler.

```
``go // Package: helixterm.io/services/terminal // File: internal/ws/hub.go
package ws
import ("digital.vasic.concurrency/pkg/workpool" "sync")
type Hub struct { sessions sync.Map // session_id → Session broadcast chan
BroadcastMsg register chan Session unregister chan *Session pool
workpool.WorkPool }
func (h Hub) Run(ctx context.Context) { for { select { case <-ctx.Done(): return case
session := <-h.register: h.sessions.Store(session.ID, session)
metrics.ActiveSessions.Inc() case session := <-h.unregister:
h.sessions.Delete(session.ID) close(session.send) metrics.ActiveSessions.Dec() case
msg := <-h.broadcast: h.sessions.Range(func(key, value any) bool { session := value.
(Session) if session.CollabRoomID == msg.RoomID { select { case session.send <-
msg.Data: default: // Slow consumer — drop message and close
h.sessions.Delete(key) close(session.send) } } return true }) } } } ``
```

### ### 2. Kafka for Cross-Pod Fan-out

When a terminal session has collaborators on different pods (because the collab fan-out needs to reach multiple pods), the Terminal Service publishes output to Kafka `helix.terminal.output`. All pods in the consumer group receive the event and forward to their local sessions. A session-to-pod routing table in Redis directs direct sends.

User A (Pod 1) —input→ SSH Proxy SSH Proxy —output→ Terminal Pod 1 →  
Kafka `helix.terminal.output` Kafka → Terminal Pod 1 (User B, observer) →  
WebSocket to User B Kafka → Terminal Pod 2 (User C, observer) → WebSocket  
to User C

### ### 3. Connection Load Balancing

The Ingress uses `nginx.ingress.kubernetes.io/affinity: cookie` for WebSocket connections to ensure sticky sessions — a user's WebSocket reconnect goes to the same pod. This avoids re-establishing SSH connections.

### ## 8.5 Circuit Breaker Thresholds

See §2.8 for the code. Summary of production thresholds:

```
yaml # Per-service circuit breaker configuration in ConfigMap
circuit_breakers: auth-service: max_requests_half_open: 3 interval: 10s
timeout: 30s trip_failure_ratio: 0.5 min_requests: 10 vault-service:
```

---

## # 7. API Specification

---

```
max_requests_half_open: 3 interval: 10s timeout: 30s trip_failure_ratio:
0.5 min_requests: 10 ssh-proxy: max_requests_half_open: 2 interval: 30s
timeout: 60s trip_failure_ratio: 0.3 # Trip earlier – SSH failures are
critical min_requests: 5 ai-service: max_requests_half_open: 5 interval:
10s timeout: 60s trip_failure_ratio: 0.7 # AI can have higher failure
tolerance min_requests: 20
```

### ## 8.6 SSH Connection Pooling

The SSH Proxy maintains a connection pool of authenticated SSH connections per (user\_id, host\_id) tuple to avoid the overhead of re-establishing the TCP + SSH handshake for every new session:

```
```go // Package: helixterm.io/services/ssh-proxy // File: internal/pool/ssh_pool.go
package pool
import ( "golang.org/x/crypto/ssh" "sync" "time" )
type SSHConnectionPool struct { mu sync.Mutex pools map[string]*HostPool //
key: "user_id:host_id" }
type HostPool struct { connections []*pooledConn maxIdle int maxActive int
idleTimeout time.Duration }
type pooledConn struct { client *ssh.Client lastUsed time.Time inUse bool }
// Get returns an existing idle connection or creates a new one. // Connections are
reused across sessions to amortize SSH handshake cost. func (p
SSHConnectionPool) Get(userID, hostID string) (ssh.Client, error) { key := userID +
":" + hostID p.mu.Lock() pool, ok := p.pools[key] if !ok { pool =
&HostPool{maxIdle: 5, maxActive: 20, idleTimeout: 10 * time.Minute}
p.pools[key] = pool } p.mu.Unlock() return pool.get() } ```
```

8.7 Kafka Throughput Optimization

- **Producer batching:** batch.size=1MB, linger.ms=5 — accumulate messages for 5ms before sending, enabling batches of up to 1MB. - **Snappy compression:** reduces network bandwidth by ~60% for JSON payloads. - **Async producer mode:** terminal output events use async fire-and-forget producers; audit events use synchronous producers with acks=all. - **Partition key design:** session_id for terminal events ensures all events from a session land in the same partition, preserving order.

```
```go // Package: digital.vasic.messaging/pkg/kafka // Producer configuration for
high-throughput event topics
var highThroughputProducerConfig = sarama.NewConfig() func init()
{ highThroughputProducerConfig.Producer.Return.Successes = false // Async
highThroughputProducerConfig.Producer.Return.Errors = true
highThroughputProducerConfig.Producer.Compression =
sarama.CompressionSnappy
```

---

## # 7. API Specification

---

```
highThroughputProducerConfig.Producer.Flush.Bytes = 1 * 1024 * 1024 // 1MB
highThroughputProducerConfig.Producer.Flush.Frequency = 5 *
time.Millisecond highThroughputProducerConfig.Producer.RequiredAcks =
sarama.WaitForLocal // acks=1
highThroughputProducerConfig.Net.MaxOpenRequests = 10 }

// Producer configuration for audit events (durability critical) var
auditProducerConfig = sarama.NewConfig() func init()
{ auditProducerConfig.Producer.Return.Successes = true // Sync
auditProducerConfig.Producer.Compression = sarama.CompressionSnappy
auditProducerConfig.Producer.RequiredAcks = sarama.WaitForAll // acks=all
auditProducerConfig.Net.MaxOpenRequests = 5
auditProducerConfig.Producer.Idempotent = true // Exactly-once } ``
```

### ## 8.8 CDN Strategy

All Flutter Web SPA static assets (compiled JS, CSS, fonts, images) are served via CloudFront (AWS deployment) or Google Cloud CDN (GCP deployment) with: - **Cache-Control:** public, max-age=31536000, immutable (content-addressed filenames). - **Brotli compression** for all text assets. - **HTTP/2 push** for critical JS chunks. - **Edge locations:** 50+ PoPs globally. - **API Gateway:** NOT behind CDN — all API requests go directly to the Kubernetes Ingress. CDN is for static assets only.

### ## 8.9 Load Balancing Algorithms

| Layer | Algorithm | Tool | |—|—|—| | External (Ingress) | Round-robin (NGINX default) | Nginx Ingress Controller | | WebSocket (Ingress) | Cookie-based sticky session | NGINX sticky annotation | | gRPC (internal) | Round-robin (headless service DNS) | Envoy (Istio) | | PostgreSQL reads | Round-robin across replicas | PgBouncer + custom routing in digital.vasic.database | | Redis | Consistent hashing (keyslot) | Redis Cluster | | Kafka consumers | Partition assignment (cooperative-sticky) | Kafka consumer group protocol |

### ## 8.10 Observability Stack

`` Services → OpenTelemetry SDK (digital.vasic.observability) → OTel Collector (DaemonSet) |—— Traces → Jaeger (Tempo in cloud deployments) |—— Metrics → Prometheus → Grafana |—— Logs → Loki → Grafana

Prometheus scrapes: - All service /metrics endpoints (Gin middleware injects metrics) - Node exporter (node CPU/mem/disk) - Kafka exporter (JMX metrics) - PostgreSQL exporter - Redis exporter - RabbitMQ management plugin

Key Dashboards (Grafana): - Platform Overview (RPS, error rates, p50/p95/p99 latency per service) - SSH Session Health (active sessions, connection establishment time, error rates) - Kafka Consumer Lag (per consumer group, per topic) - Security Dashboard (login anomalies, failed auth rates, anomaly detections) - SLO Dashboard (error budget burn rate per service) ``

---

## # 7. API Specification

---

```
``go // Package: digital.vasic.observability/pkg/tracing // Every handler
automatically wraps requests with OTEL spans
import ("go.opentelemetry.io/otel" "go.opentelemetry.io/otel/trace"
"go.opentelemetry.io/otel/exporters/otlp/otlptrace/otlptracegrpc")
func InitTracer(serviceName, endpoint string) (trace.TracerProvider, error)
{ exporter, err := otlptracegrpc.New(context.Background(),
otlptracegrpc.WithEndpoint(endpoint),
otlptracegrpc.WithTLSClientConfig(&tls.Config{...}),) if err != nil { return nil, err }
tp := sdktrace.NewTracerProvider(sdktrace.WithBatcher(exporter),
sdktrace.WithResource(resource.NewWithAttributes(semconv.SchemaURL,
semconv.ServiceNameKey.String(serviceName),
attribute.String("service.version", version.Version),
attribute.String("deployment.environment", os.Getenv("ENV")),)),
sdktrace.WithSampler(sdktrace.ParentBased(sdktrace.TraceIDRatioBased(0.1))), //
10% sampling) otel.SetTracerProvider(tp) return tp, nil } ``
```

---

# 9. Kubernetes Deployment

## 9.1 Namespace Organization

helixterm-prod	# Production workloads
helixterm-staging	# Staging workloads
helixterm-dev	# Development workloads
helixterm-infra	# Infrastructure: Kafka, RabbitMQ, Redis,
PostgreSQL operators	
helixterm-monitoring	# Prometheus, Grafana, Jaeger, Loki,
Alertmanager	
helixterm-security	# SPIRE, cert-manager, external-secrets-
operator	
helixterm-ingress	# Nginx Ingress Controller

## 9.2 Example Service Kubernetes Resources: Auth Service

### Deployment

```
deploy/kubernetes/helixterm-prod/auth-service/deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
 name: auth-service
 namespace: helixterm-prod
 labels:
 app: auth-service
 version: "1.0.0"
 app.kubernetes.io/name: auth-service
 app.kubernetes.io/part-of: helixterm
spec:
 replicas: 5
 selector:
 matchLabels:
 app: auth-service
 strategy:
 type: RollingUpdate
 rollingUpdate:
 maxSurge: 2
 maxUnavailable: 0
 template:
 metadata:
 labels:
 app: auth-service
 version: "1.0.0"
 annotations:
 prometheus.io/scrape: "true"
 prometheus.io/port: "9090"
 prometheus.io/path: "/metrics"
 sidecar.istio.io/inject: "true"
 spec:
 serviceAccountName: auth-service
 terminationGracePeriodSeconds: 30
 affinity:
 podAntiAffinity:
 preferredDuringSchedulingIgnoredDuringExecution:
 - weight: 100
```

```

 podAffinityTerm:
 labelSelector:
 matchLabels:
 app: auth-service
 topologyKey: kubernetes.io/hostname
- weight: 50
 podAffinityTerm:
 labelSelector:
 matchLabels:
 app: auth-service
 topologyKey: topology.kubernetes.io/zone
containers:
- name: auth-service
 image: registry.helixterm.io/helixterm/auth-service:1.0.0
 ports:
 - name: http
 containerPort: 8081
 protocol: TCP
 - name: grpc
 containerPort: 9081
 protocol: TCP
 - name: metrics
 containerPort: 9090
 protocol: TCP
 env:
 - name: ENV
 value: production
 - name: PORT
 value: "8081"
 - name: GRPC_PORT
 value: "9081"
 - name: DATABASE_URL
 valueFrom:
 secretKeyRef:
 name: auth-service-secrets
 key: database_url
 - name: REDIS_ADDRS
 valueFrom:
 secretKeyRef:
 name: auth-service-secrets
 key: redis_addrs
 - name: KAFKA_BROKERS
 valueFrom:
 configMapKeyRef:

```

```
 name: kafka-config
 key: brokers
- name: RABBITMQ_URL
 valueFrom:
 secretKeyRef:
 name: rabbitmq-secrets
 key: url
- name: OTEL_EXPORTER_OTLP_ENDPOINT
 value: "otel-collector.helixterm-monitoring.svc:4317"
- name: SERVICE_NAME
 value: auth-service
resources:
 requests:
 memory: "256Mi"
 cpu: "250m"
 limits:
 memory: "512Mi"
 cpu: "1000m"
readinessProbe:
 httpGet:
 path: /health/ready
 port: 8081
 initialDelaySeconds: 5
 periodSeconds: 10
 successThreshold: 1
 failureThreshold: 3
 timeoutSeconds: 5
livenessProbe:
 httpGet:
 path: /health/live
 port: 8081
 initialDelaySeconds: 10
 periodSeconds: 30
 failureThreshold: 3
 timeoutSeconds: 5
startupProbe:
 httpGet:
 path: /health/startup
 port: 8081
 failureThreshold: 30
 periodSeconds: 5
volumeMounts:
- name: tmp
 mountPath: /tmp
```

```
volumes:
 - name: tmp
 emptyDir: {}
securityContext:
 runAsNonRoot: true
 runAsUser: 65532
 runAsGroup: 65532
 fsGroup: 65532
 seccompProfile:
 type: RuntimeDefault
```

## Service

```
deploy/kubernetes/helixterm-prod/auth-service/service.yaml
apiVersion: v1
kind: Service
metadata:
 name: auth-service
 namespace: helixterm-prod
 labels:
 app: auth-service
spec:
 selector:
 app: auth-service
 ports:
 - name: http
 port: 8081
 targetPort: 8081
 protocol: TCP
 - name: grpc
 port: 9081
 targetPort: 9081
 protocol: TCP
 type: ClusterIP
```

## HPA (Horizontal Pod Autoscaler)

```
deploy/kubernetes/helixterm-prod/auth-service/hpa.yaml
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
 name: auth-service
 namespace: helixterm-prod
spec:
```

```
scaleTargetRef:
 apiVersion: apps/v1
 kind: Deployment
 name: auth-service
minReplicas: 5
maxReplicas: 50
metrics:
- type: Resource
 resource:
 name: cpu
 target:
 type: Utilization
 averageUtilization: 70
- type: Resource
 resource:
 name: memory
 target:
 type: Utilization
 averageUtilization: 80
- type: Pods
 pods:
 metric:
 name: http_requests_per_second
 target:
 type: AverageValue
 averageValue: "2000"
behavior:
 scaleDown:
 stabilizationWindowSeconds: 300
 policies:
 - type: Percent
 value: 10
 periodSeconds: 60
 scaleUp:
 stabilizationWindowSeconds: 30
 policies:
 - type: Percent
 value: 100
 periodSeconds: 30
 - type: Pods
 value: 5
 periodSeconds: 30
 selectPolicy: Max
```

## PodDisruptionBudget

```
deploy/kubernetes/helixterm-prod/auth-service/pdb.yaml
apiVersion: policy/v1
kind: PodDisruptionBudget
metadata:
 name: auth-service
 namespace: helixterm-prod
spec:
 minAvailable: 3
 selector:
 matchLabels:
 app: auth-service
```

## ConfigMap

```
deploy/kubernetes/helixterm-prod/auth-service/configmap.yaml
apiVersion: v1
kind: ConfigMap
metadata:
 name: auth-service-config
 namespace: helixterm-prod
data:
 access_token_ttl: "900"
 refresh_token_ttl: "2592000"
 max_sessions_per_user: "20"
 mfa_challenge_ttl: "300"
 password_min_length: "12"
 argon2_time_cost: "3"
 argon2_memory_cost: "65536"
 argon2_parallelism: "4"
 login_rate_limit_attempts: "5"
 login_rate_limit_window_s: "900"
 jwks_cache_ttl: "300"
 log_level: "info"
 log_format: "json"
```

## ServiceAccount

```
deploy/kubernetes/helixterm-prod/auth-service/
 serviceaccount.yaml
apiVersion: v1
kind: ServiceAccount
metadata:
```

```
name: auth-service
namespace: helixterm-prod
annotations:
 eks.amazonaws.com/role-arn: arn:aws:iam::ACCOUNT_ID:role/
 helixterm-auth-service
```

## 9.3 SSH Proxy Service (High-Connection-Count Specialization)

```
deploy/kubernetes/helixterm-prod/ssh-proxy/deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
 name: ssh-proxy
 namespace: helixterm-prod
spec:
 replicas: 20
 template:
 metadata:
 annotations:
 sidecar.istio.io/inject: "true"
 sidecar.istio.io/proxyCPU: "500m"
 sidecar.istio.io/proxyMemory: "256Mi"
 spec:
 serviceAccountName: ssh-proxy
 affinity:
 podAntiAffinity:
 requiredDuringSchedulingIgnoredDuringExecution:
 - labelSelector:
 matchLabels:
 app: ssh-proxy
 topologyKey: kubernetes.io/hostname # Strict: one
 pod per node
 containers:
 - name: ssh-proxy
 image: registry.helixterm.io/helixterm/ssh-proxy:1.0.0
 ports:
 - containerPort: 8090
 resources:
 requests:
 memory: "1Gi"
 cpu: "1000m"
 limits:
```

```

 memory: "4Gi"
 cpu: "4000m"
 env:
 - name: MAX_CONNECTIONS
 value: "5000"
 - name: CONNECTION_TIMEOUT
 value: "30"
 - name: KEEPALIVE_INTERVAL
 value: "30"
 - name: POOL_MAX_IDLE
 value: "5"
 - name: POOL_IDLE_TIMEOUT
 value: "600"
 readinessProbe:
 httpGet:
 path: /health/ready
 port: 8090
 initialDelaySeconds: 5
 periodSeconds: 10
 livenessProbe:
 httpGet:
 path: /health/live
 port: 8090
 initialDelaySeconds: 10
 periodSeconds: 30
 # Increase kernel connection limits
 initContainers:
 - name: sysctl
 image: busybox:1.36
 command: ['sh', '-c', 'sysctl -w net.core.somaxconn=65535
net.ipv4.tcp_max_syn_backlog=65535']
 securityContext:
 privileged: true
 securityContext:
 sysctl:
 - name: net.ipv4.ip_local_port_range
 value: "1024 65535"
 - name: net.core.somaxconn
 value: "65535"

```

## 9.4 Terminal Service (WebSocket at Scale)

```

apiVersion: apps/v1
kind: Deployment

```

```

metadata:
 name: terminal-service
 namespace: helixterm-prod
spec:
 replicas: 50
 template:
 spec:
 containers:
 - name: terminal-service
 image: registry.helixterm.io/helixterm/terminal-
 service:1.0.0
 resources:
 requests:
 memory: "512Mi"
 cpu: "500m"
 limits:
 memory: "2Gi"
 cpu: "2000m"
 env:
 - name: MAX_WEBSOCKET_CONNECTIONS
 value: "5000"
 - name: SCROLLBACK_LINES
 value: "10000"
 - name: IDLE_TIMEOUT_S
 value: "1800"
 - name: BUFFER_SIZE
 value: "65536"

```

## 9.5 Ingress Configuration

```

deploy/kubernetes/helixterm-prod/ingress/ingress.yaml
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
 name: helixterm-ingress
 namespace: helixterm-prod
 annotations:
 nginx.ingress.kubernetes.io/rewrite-target: /
 nginx.ingress.kubernetes.io/proxy-body-size: 100m
 nginx.ingress.kubernetes.io/proxy-read-timeout: "3600" #
 WebSocket keep-alive
 nginx.ingress.kubernetes.io/proxy-send-timeout: "3600"
 nginx.ingress.kubernetes.io/proxy-connect-timeout: "10"
 nginx.ingress.kubernetes.io/configuration-snippet: |

```

```

 more_set_headers "X-Frame-Options: DENY";
 more_set_headers "X-Content-Type-Options: nosniff";
 more_set_headers "X-XSS-Protection: 1; mode=block";
 more_set_headers "Referrer-Policy: strict-origin-when-cross-
 origin";
 more_set_headers "Strict-Transport-Security: max-
 age=31536000; includeSubDomains; preload";
 more_set_headers "Content-Security-Policy: default-src
 'self'; connect-src 'self' wss://api.helixterm.io";
 nginx.ingress.kubernetes.io/use-regex: "true"
 nginx.ingress.kubernetes.io/affinity: "cookie"
 nginx.ingress.kubernetes.io/session-cookie-name:
 "HELIX_AFFINITY"
 nginx.ingress.kubernetes.io/session-cookie-expires: "172800"
 cert-manager.io/cluster-issuer: "letsencrypt-prod"
spec:
 ingressClassName: nginx
 tls:
 - hosts:
 - api.helixterm.io
 - app.helixterm.io
 secretName: helixterm-tls
 rules:
 - host: api.helixterm.io
 http:
 paths:
 - path: /api/v1
 pathType: Prefix
 backend:
 service:
 name: gateway-service
 port:
 number: 8080
 - path: /ws/v1
 pathType: Prefix
 backend:
 service:
 name: gateway-service
 port:
 number: 8080
 - host: app.helixterm.io
 http:
 paths:
 - path: /
 pathType: Prefix
 backend:

```

```

service:
 name: flutter-web-spa
 port:
 number: 80

```

## 9.6 Helm Chart Structure

```

deploy/helm/helixterm/
├─ Chart.yaml
├─ values.yaml
├─ values-prod.yaml
├─ values-staging.yaml
├─ values-dev.yaml
├─ templates/
│ └─ _helpers.tpl
│ └─ NOTES.txt
│ └─ namespace.yaml
│ └─ serviceaccount.yaml
│ └─ configmap.yaml
│ └─ secrets.yaml # Only schema – actual secrets
from ESO
├─ gateway/
│ └─ deployment.yaml
│ └─ service.yaml
│ └─ hpa.yaml
│ └─ pdb.yaml
├─ auth/
│ └─ deployment.yaml
│ └─ service.yaml
│ └─ hpa.yaml
│ └─ pdb.yaml
├─ ... (one directory per service)
├─ networking/
│ └─ ingress.yaml
│ └─ network-policies.yaml
│ └─ istio-policies.yaml
├─ rbac/
│ └─ roles.yaml
│ └─ rolebindings.yaml
└─ charts/
 └─ postgresql/ # Bitnami PostgreSQL (per-service
override)

```

```

└─ redis/ # Bitnami Redis Cluster
└─ kafka/ # Strimzi Kafka operator dependency

deploy/helm/helixterm/Chart.yaml
apiVersion: v2
name: helixterm
description: HelixTerminator – Next-Generation Enterprise SSH
 Client Platform
type: application
version: 1.0.0
appVersion: "1.0.0"
maintainers:
- name: HelixDevelopment
 url: https://github.com/HelixDevelopment
dependencies:
- name: postgresql
 version: "~15.0"
 repository: https://charts.bitnami.com/bitnami
 condition: postgresql.enabled
- name: redis
 version: "~20.0"
 repository: https://charts.bitnami.com/bitnami
 condition: redis.enabled

```

## 9.7 Secret Management (External Secrets Operator)

```

deploy/kubernetes/helixterm-prod/secrets/external-secret-
 auth.yaml
apiVersion: external-secrets.io/v1beta1
kind: ExternalSecret
metadata:
 name: auth-service-secrets
 namespace: helixterm-prod
spec:
 refreshInterval: 5m
 secretStoreRef:
 name: aws-secrets-manager
 kind: ClusterSecretStore
 target:
 name: auth-service-secrets
 creationPolicy: Owner
 data:
 - secretKey: database_url

```

```

remoteRef:
 key: helixterm/prod/auth-service
 property: database_url
- secretKey: redis_addrs
remoteRef:
 key: helixterm/prod/auth-service
 property: redis_addrs
- secretKey: jwt_private_key
remoteRef:
 key: helixterm/prod/auth-service
 property: jwt_private_key

```

## 9.8 Pod Resource Limits Summary

Service	CPU Request	CPU Limit	Mem Request	Mem Limit
Gateway	500m	2000m	512Mi	1Gi
Auth	250m	1000m	256Mi	512Mi
User	200m	500m	256Mi	512Mi
Vault	250m	500m	256Mi	512Mi
Host	200m	500m	256Mi	512Mi
SSH Proxy	1000m	4000m	1Gi	4Gi
Terminal	500m	2000m	512Mi	2Gi
SFTP	500m	2000m	512Mi	2Gi
Port Forward	250m	1000m	256Mi	1Gi
Collab	500m	2000m	512Mi	2Gi
Audit	250m	1000m	256Mi	1Gi
Analytics	500m	2000m	512Mi	2Gi
AI	1000m	4000m	1Gi	8Gi
Recording	500m	2000m	512Mi	2Gi
PKI	250m	500m	256Mi	512Mi
Org	200m	500m	256Mi	512Mi
Billing	200m	500m	256Mi	512Mi
Notification	200m	500m	256Mi	512Mi
Config	100m	250m	128Mi	256Mi
Health	100m	250m	128Mi	256Mi
	250m	1000m	256Mi	1Gi

Service	CPU Request	CPU Limit	Mem Request	Mem Limit
Container Bridge				
HelixTrack Bridge	200m	500m	256Mi	512Mi

## 9.9 Rollout Strategy (Zero-Downtime Deploys)

All services use the following rollout process:

1. **Canary phase:** Argo Rollouts with 5% traffic to new version.
2. **Metrics gate:** Automatic promotion if error rate < 0.1% and p99 latency < 200ms over 10 minutes.
3. **Full rollout:** Progressive traffic increase: 5% → 25% → 50% → 100% over 30 minutes.
4. **Automatic rollback:** If metrics gate fails at any step, Argo Rollouts reverts to previous version.

```
deploy/kubernetes/helixterm-prod/auth-service/rollout.yaml
apiVersion: argoproj.io/v1alpha1
kind: Rollout
metadata:
 name: auth-service
 namespace: helixterm-prod
spec:
 replicas: 5
 strategy:
 canary:
 canaryService: auth-service-canary
 stableService: auth-service-stable
 trafficRouting:
 istio:
 virtualService:
 name: auth-service-vs
 destinationRule:
 name: auth-service-ds
 canarySubsetName: canary
 stableSubsetName: stable
 steps:
 - setWeight: 5
```

- pause: {duration: 2m}
- analysis:
  - templates:
    - templateName: error-rate
    - templateName: latency-p99
- setWeight: 25
- pause: {duration: 5m}
- setWeight: 50
- pause: {duration: 5m}
- setWeight: 100

---

## # 10. Submodule Integration

### ## 10.1 Integration Philosophy (HelixConstitution §11.4.74)

Per HelixConstitution §11.4.74 (submodule-catalogue-first discovery, extend-don't-reimplement):

- **Never reimplement** functionality that exists in a submodule. If `digital.vasic.security` provides AES-256-GCM encryption, use it — do not write a new encryption package. - **Always check** `submodules-catalogue.md` before adding a new dependency. - **All containerized workloads** must use `digital.vasic.containers` per §11.4.76. - **All spec documents** must be registered with `digital.vasic.docs_chain` per §11.4.73.

> **Canonical reference:** `10_submodule_integration.md` is the authoritative submodule-integration document (versions, import-path convention, service registry). §10.2–§10.13 below are this document's own architectural walkthrough of the same submodules and are kept for context, but where a version number or module-path line below conflicts with `10_submodule_integration.md`, that document — reconciled against `CANONICAL_FACTS.md` — wins. Per `CANONICAL_FACTS.md`, the Go import-path rename (dot-form `digital.vasic.*` → slash-form `digital.vasic/*`) is DEFERRED, high-churn (600+ refs), cross-document work; the `digital.vasic.*`-dotted import paths inside the code samples below are therefore left as-is and flagged rather than mass-rewritten.

### ## 10.2 `digital.vasic.containers`

**Version used:** see `10_submodule_integration.md` (canonical: v1.0.5) — this section's version reference is superseded per `CANONICAL_FACTS.md`. **Module path:** `digital.vasic.containers` (dot-form; DEFERRED rename to slash-form per `CANONICAL_FACTS.md` — see `10_submodule_integration.md` for the

canonical import-path convention note) **Used by:** SSH Proxy Service, Container Bridge Service

### ### ContainerRuntime Abstraction

digital.vasic.containers provides a unified ContainerRuntime interface that abstracts Docker, Podman, and Kubernetes Pod exec, enabling container-native terminal sessions without an SSH daemon:

```
```go // Package: digital.vasic.containers/pkg/runtime // Interface: ContainerRuntime
```

```
package runtime
```

```
import "context"
```

```
// ContainerRuntime is the unified abstraction over Docker, Podman, and Kubernetes. // Per HelixConstitution §11.4.76, ALL container interactions in HelixTerminator // MUST go through this interface. type ContainerRuntime interface { // Exec starts a process inside a running container. Exec(ctx context.Context, req ExecRequest) (ExecConnection, error)
```

```
// ListContainers lists containers matching the filter. ListContainers(ctx context.Context, filter ContainerFilter) ([]Container, error)
```

```
// InspectContainer returns detailed container/pod information. InspectContainer(ctx context.Context, id string) (*ContainerInfo, error)
```

```
// Logs streams container logs. Logs(ctx context.Context, id string, opts LogOptions) (io.ReadCloser, error)
```

```
// Stats streams container resource statistics. Stats(ctx context.Context, id string) (<-chan ContainerStats, error)
```

```
// Start, Stop, Restart, Remove — lifecycle operations Start(ctx context.Context, id string) error Stop(ctx context.Context, id string, timeout time.Duration) error Restart(ctx context.Context, id string, timeout time.Duration) error Remove(ctx context.Context, id string, force bool) error
```

```
// Health returns the runtime's connection health. Health(ctx context.Context) error } ```
```

Runtime Registration in SSH Proxy

```
```go // Package: helixterm.io/services/ssh-proxy // File: internal/runtime/registry.go
```

```
package runtime
```

```

import ("digital.vasic.containers/pkg/runtime"
"digital.vasic.containers/pkg/docker" "digital.vasic.containers/
pkg/podman" "digital.vasic.containers/pkg/kubernetes")

type RuntimeRegistry struct { runtimes
map[string]runtime.ContainerRuntime }

func NewRegistry(cfg Config) (*RuntimeRegistry, error) { reg :=
&RuntimeRegistry{runtimes:
make(map[string]runtime.ContainerRuntime)}

if cfg.DockerEnabled { dockerRT, err :=
docker.New(docker.Config{ Host: cfg.DockerHost, // e.g., "unix:///
var/run/docker.sock" Version: "1.47", }) if err != nil { return nil,
fmt.Errorf("docker runtime: %w", err) } reg.runtimes["docker"]
= dockerRT }

if cfg.PodmanEnabled { podmanRT, err :=
podman.New(podman.Config{ Host: cfg.PodmanHost, // e.g.,
"unix:///run/user/1000/podman/podman.sock" }) if err != nil
{ return nil, fmt.Errorf("podman runtime: %w", err) }
reg.runtimes["podman"] = podmanRT }

if cfg.KubernetesEnabled { k8sRT, err :=
kubernetes.New(kubernetes.Config{ Kubeconfig:
cfg.Kubeconfig, InCluster: cfg.K8sInCluster, }) if err != nil
{ return nil, fmt.Errorf("kubernetes runtime: %w", err) }
reg.runtimes["kubernetes"] = k8sRT }

return reg, nil }

func (r *RuntimeRegistry) Get(runtimeType string)
(runtime.ContainerRuntime, bool) { rt, ok :=
r.runtimes[runtimeType] return rt, ok } ``

Health Monitoring via digital.vasic.containers
``go // Package: helixterm.io/services/container-bridge // File:
internal/health/watcher.go // Uses digital.vasic.containers health
checks + lifecycle monitoring

package health

import ("digital.vasic.containers/pkg/runtime"
"digital.vasic.containers/pkg/lifecycle" "digital.vasic.containers/
pkg/monitor" "helixterm.io/services/container-bridge/internal/
events")

type ContainerHealthWatcher struct { runtime
runtime.ContainerRuntime monitor monitor.Monitor producer
events.Producer }

```

```

func (w *ContainerHealthWatcher) Start(ctx context.Context) { //
Subscribe to container lifecycle events from
digital.vasic.containers eventCh := w.monitor.Subscribe(ctx,
monitor.EventFilter{ Types: []monitor.EventType{
monitor.EventContainerStart, monitor.EventContainerStop,
monitor.EventContainerDie,
monitor.EventContainerHealthStatus,
monitor.EventPodPhaseChange, }, })

for { select { case <-ctx.Done(): return case evt := <-eventCh: //
Forward as Kafka helix.containers.health event
w.producer.Produce(ctx,
events.ContainerHealthEvent{ ClusterID: w.clusterID,
ContainerID: evt.ContainerID, ContainerName:
evt.ContainerName, EventType: string(evt.Type), Status:
evt.Status, OccurredAt: evt.Timestamp, }) } } } ``

10.3 digital.vasic.security

```

**Version used:** see 10\_submodule\_integration.md (canonical: v1.4.2) — this section’s version reference is superseded per CANONICAL\_FACTS.md. **Module path:** digital.vasic.security (dot-form; DEFERRED rename to slash-form per CANONICAL\_FACTS.md — see 10\_submodule\_integration.md for the canonical import-path convention note) **Used by:** Vault Service, Keychain Service, Auth Service, PKI Service

### Encryption Primitives

```

``go // Package: digital.vasic.security/pkg/aes // Provides:
AES-256-GCM encryption, secure random generation, key
derivation

import "digital.vasic.security/pkg/aes"

encryptor := aes.NewGCMEncryptor() ciphertext, nonce, err :=
encryptor.Encrypt(plaintextBytes, keyBytes) plaintext, err :=
encryptor.Decrypt(ciphertext, nonce, keyBytes)

// Package: digital.vasic.security/pkg/storage // Provides: secure
key storage (OS-level hardware-backed where available)

import "digital.vasic.security/pkg/storage"

store := storage.New(storage.Config{ Backend:
storage.BackendSecureEnclave, // or: DPAPI, AndroidKeystore,
LinuxKeyring, SoftwareKMS Namespace: "helixterm.vault", })
err := store.Store("key_id_001", privateKeyBytes) keyBytes, err :=
store.Load("key_id_001") ``

Usage in Vault Service

```

Every call to create or update a vault item in the Vault Service passes through `digital.vasic.security`:

```
```go // Package: helixterm.io/services/vault // File: internal/
service/vault_service.go

import ( "digital.vasic.security/pkg/aes" "digital.vasic.security/
pkg/kdf" )

func (s *VaultService) ReEncryptForMigration(ctx
context.Context, item domain.VaultItem, newKeyID string) error
{ // Existing item is already encrypted — this is used for key
rotation // Server never has the plaintext; it only re-wraps the
encryption key newKeyMaterial, err :=
s.keychain.GetWrappedKey(ctx, newKeyID) if err != nil { return
err } // The re-wrapping is done client-side; server stores the
new ciphertext from client return s.repo.UpdateKeyID(ctx,
item.ID, newKeyID) } ```
```

Argon2id Password Hashing (Auth Service)

```
```go // Package: digital.vasic.security/pkg/argon2

import "digital.vasic.security/pkg/argon2"

hasher := argon2.NewHasher(argon2.Config{ TimeCost: 3,
MemoryCost: 65536, // 64 MiB Parallelism: 4, KeyLen: 32,
SaltLen: 16, })

hash, err := hasher.Hash(password) valid, err :=
hasher.Verify(password, hash) ```
```

## 10.4 `digital.vasic.auth`

**Version used:** see `10_submodule_integration.md` (canonical: v2.3.1) — this section’s version reference is superseded per `CANONICAL_FACTS.md`. **Module path:** `digital.vasic.auth` (dot-form; DEFERRED rename to slash-form per `CANONICAL_FACTS.md` — see `10_submodule_integration.md` for the canonical import-path convention note) **Used by:** Auth Service, HelixTrack Bridge Service, Billing Service (Stripe OAuth)

### OAuth2 Flows

```
```go // Package: digital.vasic.auth/pkg/oauth2 // Used in
HelixTrack Bridge for OAuth2 integration with helixtrack.ru/
core

import "digital.vasic.auth/pkg/oauth2"

helixtrackOAuth2 := oauth2.NewClient(oauth2.Config{ ClientID:
os.Getenv("HELIXTRACK_CLIENT_ID"), ClientSecret:
os.Getenv("HELIXTRACK_CLIENT_SECRET"), Scopes:
[]string{"issues:read", "projects:read", "sprints:read"}, AuthURL:
```

```

“https://helixtrack.ru/oauth2/authorize”, TokenURL: “https://
helixtrack.ru/oauth2/token”, RedirectURL: “https://
api.helixterm.io/api/v1/helixtrack/oauth/callback”, })

// Authorization code flow authURL :=
helixtrackOAuth2.AuthCodeURL(state,
oauth2.AccessTypeOffline)

// Exchange code for token token, err :=
helixtrackOAuth2.Exchange(ctx, code)

// Token management (auto-refresh, storage) tokenManager :=
oauth2.NewTokenManager(oauth2.TokenManagerConfig{ Client:
helixtrackOAuth2, Storage: tokenStorage, // PostgreSQL-backed
storage }) ``

### JWT Management (Auth Service)

``go // Package: digital.vasic.auth/pkg/jwt // Used in Auth Service
for token issuance and validation

import “digital.vasic.auth/pkg/jwt”

manager := jwt.NewManager(jwt.Config{ Algorithm: jwt.EdDSA,
Issuer: “https://auth.helixterm.io”, Audience: []string{“https://
api.helixterm.io”}, PrivateKeyID: “helix-2026-06-key-01”, })

// Issue token, err := manager.Issue(jwt.Claims{ Subject: userID,
ExpiresAt: time.Now().Add(15 * time.Minute), // ... other
claims })

// Validate claims, err := manager.Validate(tokenString) ``

## 10.5 digital.vasic.messaging

Version used: see 10_submodule_integration.md (canonical:
v1.5.3) — this section’s version reference is superseded per
CANONICAL_FACTS.md. Module path: digital.vasic.messaging
(dot-form; DEFERRED rename to slash-form per
CANONICAL_FACTS.md — see 10_submodule_integration.md for the
canonical import-path convention note) Used by: All services
that produce or consume Kafka topics or RabbitMQ queues

### Kafka Producer (all services)

``go // Package: digital.vasic.messaging/pkg/kafka

import “digital.vasic.messaging/pkg/kafka”

// High-throughput async producer (for analytics, terminal
output) asyncProducer, err :=
kafka.NewAsyncProducer(kafka.ProducerConfig{ Brokers:
strings.Split(os.Getenv(“KAFKA_BROKERS”), “,”), TLSConfig:
loadTLSConfig(), Compression: kafka.CompressionSnappy,

```

```

BatchSize: 1 * 1024 * 1024, Linger: 5 * time.Millisecond, Acks:
kafka.AcksLocal, })

// Durable sync producer (for audit events) syncProducer, err :=
kafka.NewSyncProducer(kafka.ProducerConfig{ Brokers:
strings.Split(os.Getenv("KAFKA_BROKERS"), ","), TLSConfig:
loadTLSConfig(), Compression: kafka.CompressionSnappy, Acks:
kafka.AcksAll, Idempotent: true, })

// Produce with partition key err =
asyncProducer.Produce(kafka.Message{ Topic:
"helix terminal.commands", Key: []byte(sessionID), // Ensure
ordering per session Value: eventJSON, Timestamp:
time.Now(), }) ``

### RabbitMQ Publisher (SSH Proxy → command queue)
``go // Package: digital.vasic.messaging/pkg/rabbitmq
import "digital.vasic.messaging/pkg/rabbitmq"

publisher, err := rabbitmq.NewPublisher(rabbitmq.Config{ URL:
os.Getenv("RABBITMQ_URL"), Exchange: "helix.commands",
ContentType: "application/json", Mandatory: true, Persistent:
true, })

err = publisher.Publish(rabbitmq.Message{ Exchange:
"helix.commands", RoutingKey: "ssh.connect", Body: cmdJSON,
Priority: 0, Expiration: "30000", // 30s TTL }) ``

## 10.6 digital.vasic.observability

Version used: see 10_submodule_integration.md (canonical:
v1.3.0) — this section's version reference is superseded per
CANONICAL_FACTS.md. Module path: digital.vasic.observability
(dot-form; DEFERRED rename to slash-form per
CANONICAL_FACTS.md — see 10_submodule_integration.md for the
canonical import-path convention note) Used by: All services

### Initialization Pattern
``go // Package: helixterm.io/services/auth // File: cmd/server/
main.go

import ( "digital.vasic.observability/pkg/tracing"
"digital.vasic.observability/pkg/metrics"
"digital.vasic.observability/pkg/logging" )

func main() { // Initialize structured logging (Zap-based) log :=
logging.New(logging.Config{ Level: os.Getenv("LOG_LEVEL"),
Format: "json", Fields: map[string]string{ "service": "auth-
service", "version": version.Version, "environment":
os.Getenv("ENV"), }, })

```

```

// Initialize OpenTelemetry tracing tp, err :=
tracing.Init(tracing.Config{ ServiceName: "auth-service",
Endpoint: os.Getenv("OTEL_EXPORTER_OTLP_ENDPOINT"),
SamplingRatio: 0.1, Propagator: tracing.PropagatorW3C, }) if err
!= nil { log.Fatal("failed to init tracing", "error", err) } defer
tp.Shutdown(context.Background())

// Initialize Prometheus metrics m :=
metrics.New(metrics.Config{ Namespace: "helixterm",
Subsystem: "auth", Port: 9090, Path: "/metrics", })

// Register standard Gin middleware from
digital.vasic.observability r := gin.New()
r.Use(tracing.GinMiddleware(tp))
r.Use(metrics.GinMiddleware(m)) } ``

### Custom Metrics (Auth Service)

``go // Package: helixterm.io/services/auth // File: internal/
metrics/auth_metrics.go

import "digital.vasic.observability/pkg/metrics"

var ( LoginAttemptsTotal =
metrics.NewCounter(metrics.CounterOpts{ Name:
"login_attempts_total", Help: "Total login attempts", Labels:
[]string{"outcome", "method"}, })

ActiveSessionsGauge =
metrics.NewGauge(metrics.GaugeOpts{ Name: "active_sessions",
Help: "Number of active user sessions", })

TokenIssuanceHistogram =
metrics.NewHistogram(metrics.HistogramOpts{ Name:
"token_issuance_duration_seconds", Help: "Time to issue access
tokens", Buckets: []float64{0.001, 0.005, 0.01, 0.025, 0.05, 0.1,
0.25, 0.5}, })

MFAChallengeOutcome =
metrics.NewCounter(metrics.CounterOpts{ Name:
"mfa_challenge_outcome_total", Help: "MFA challenge
outcomes", Labels: []string{"method", "outcome"}, }) ) ``

## 10.7 digital.vasic.ratelimiter

Version used: see 10_submodule_integration.md (canonical:
v1.0.9) — this section's version reference is superseded per
CANONICAL_FACTS.md. Used by: Gateway Service (per-endpoint),
Auth Service (login attempts), AI Service (inference rate)

``go // Package: digital.vasic.ratelimiter/pkg/limiter

import "digital.vasic.ratelimiter/pkg/limiter"

```

```
// Sliding window rate limiter backed by Redis rl :=
limiter.New(limiter.Config{ Redis: redisClient, Algorithm:
limiter.SlidingWindow, })

// Per-endpoint rate limits policies :=
map[string]limiter.Policy{ "POST /api/v1/auth/login": {Requests:
5, Window: 15 * time.Minute, Scope: limiter.ScopeIP}, "POST /
api/v1/auth/register": {Requests: 10, Window: time.Hour, Scope:
limiter.ScopeIP}, "POST /api/v1/auth/password/forgot":
{Requests: 3, Window: time.Hour, Scope: limiter.ScopeIP},
"POST /api/v1/vault/items": {Requests: 100, Window:
time.Minute, Scope: limiter.ScopeUser}, "POST /api/v1/ai/
complete": {Requests: 60, Window: time.Minute, Scope:
limiter.ScopeUser}, "GET /api/v1/": {Requests: 1000, Window:
time.Minute, Scope: limiter.ScopeUser}, "POST /api/v1/":
{Requests: 500, Window: time.Minute, Scope:
limiter.ScopeUser}, }
```

```
## 10.8 digital.vasic.recovery
```

Version used: see 10_submodule_integration.md (canonical: v1.2.1) — this section’s version reference is superseded per CANONICAL_FACTS.md. **Used by:** Gateway Service (circuit breakers), Port Forward Service (auto-reconnect), SSH Proxy (connection retry)

```
``go // Package: digital.vasic.recovery/pkg/retry // Used in Port
Forward Service for auto-reconnect
```

```
import "digital.vasic.recovery/pkg/retry"
```

```
reconnector := retry.New(retry.Config{ MaxAttempts: 0, //
Infinite retry for port forwards Backoff:
retry.ExponentialBackoff{ InitialInterval: 1 * time.Second,
MaxInterval: 60 * time.Second, Multiplier: 2.0, Jitter: 0.1, },
RetryIf: func(err error) bool { return !errors.Is(err,
context.Canceled) }, OnRetry: func(attempt int, err error)
{ log.Warn("reconnecting port forward", "attempt", attempt,
"error", err) metrics.PortForwardReconnects.Inc() }, })
err := reconnector.Do(ctx, func() error { return pf.Connect(ctx,
rule) }) ``
```

```
## 10.9 digital.vasic.cache
```

Version used: see 10_submodule_integration.md (canonical: v1.2.7) — this section’s version reference is superseded per CANONICAL_FACTS.md. **Used by:** All services for L1/L2 caching

```
``go // Package: digital.vasic.cache/pkg/cache
```

```
import "digital.vasic.cache/pkg/cache"
```

```
// Two-tier cache: L1 in-process (ristretto) + L2 Redis c :=
cache.NewTieredCache(cache.TieredConfig{ L1:
cache.L1Config{ MaxCost: 50 * 1024 * 1024, // 50MB
NumCounters: 500_000, }, L2: cache.L2Config{ Client:
redisClient, Prefix: "auth:", }, })

// Set with TTL err := c.Set(ctx, "jwks_cache", jwksJSON,
300*time.Second)

// Get (L1 hit → return; L1 miss → L2 hit → populate L1 →
return; L2 miss → fetch from DB) value, err := c.Get(ctx,
"jwks_cache")

// Invalidate (removes from both tiers) err = c.Delete(ctx,
"jwks_cache") ``

## 10.10 digital.vasic.concurrency
```

Version used: see 10_submodule_integration.md (canonical: v1.1.0) — this section’s version reference is superseded per CANONICAL_FACTS.md. **Used by:** Terminal Service (WebSocket hub), Recording Service (segment assembly), AI Service (batch inference)

```
``go // Package: digital.vasic.concurrency/pkg/workpool
import "digital.vasic.concurrency/pkg/workpool"

// Bounded goroutine pool for terminal I/O processing pool :=
workpool.New(workpool.Config{ Workers: runtime.NumCPU() *
4, QueueSize: 10000, PanicHandler: func(r interface{})
{ log.Error("panic in worker", "panic", r)
metrics.WorkerPanics.Inc() }, })

pool.Submit(func() { processTerminalOutput(sessionID, data) })

// Parallel fan-out for collab broadcast fanout :=
workpool.NewFanOut(workpool.FanOutConfig{ Workers: 100,
Timeout: 100 * time.Millisecond, }) fanout.Do(ctx,
participantIDs, func(participantID string) error { return
sendToParticipant(participantID, outputBytes) }) ``

## 10.11 digital.vasic.docs_chain
```

Version used: see 10_submodule_integration.md (canonical: v1.0.3) — this section’s version reference is superseded per CANONICAL_FACTS.md. **Module path:** digital.vasic.docs_chain (dot-form; DEFERRED rename to slash-form per CANONICAL_FACTS.md — see 10_submodule_integration.md for the canonical import-path convention note) **Used by:** CI/CD pipeline (spec document dependency validation)

digital.vasic.docs_chain is a Salsa-inspired DAG-based document dependency engine. In HelixTerminator, it ensures that changes to 01_core_architecture.md (this document) automatically trigger validation and regeneration gates on all dependent specification documents.

Document Dependency Registration

```
``go // Package: helixterm.io/core // File: docs/chain/registry.go //
Run at CI time (not at runtime)
```

```
package chain
```

```
import "digital.vasic.docs_chain/pkg/chain"
```

```
func RegisterDocChain() *chain.DocChain { dc := chain.New()
// Register all spec documents dc.Register(chain.Doc{ ID:
"01_core_architecture", Path: "docs/01_core_architecture.md",
Type: chain.DocTypeSpec, }) dc.Register(chain.Doc{ ID:
"02_client_architecture", Path: "docs/02_client_architecture.md",
Type: chain.DocTypeSpec, DependsOn:
[]string{"01_core_architecture"}, }) dc.Register(chain.Doc{ ID:
"03_security_runbook", Path: "docs/03_security_runbook.md",
Type: chain.DocTypeRunbook, DependsOn:
[]string{"01_core_architecture"}, }) dc.Register(chain.Doc{ ID:
"04_api_reference", Path: "docs/04_api_reference.md", Type:
chain.DocTypeReference, DependsOn:
[]string{"01_core_architecture"}, }) dc.Register(chain.Doc{ ID:
"05_deployment_guide", Path: "docs/05_deployment_guide.md",
Type: chain.DocTypeGuide, DependsOn:
[]string{"01_core_architecture", "04_api_reference"}, })
dc.Register(chain.Doc{ ID: "06_database_migrations", Path:
"docs/06_database_migrations.md", Type: chain.DocTypeSpec,
DependsOn: []string{"01_core_architecture"}, })
return dc } ``
```

CI/CD Integration

```
yaml # .github/workflows/docs-chain.yml name: Docs Chain
Validation on: pull_request: paths: - 'docs/**' jobs:
validate: runs-on: ubuntu-latest steps: - uses: actions/
checkout@v4 - name: Install docs-chain CLI run: go install
digital.vasic.docs_chain/cmd/docschain@v0.1.5 - name:
Validate doc dependency graph run: docschain validate --
config docs/chain/registry.go - name: Check for stale
dependents run: docschain check-stale --changed docs/
01_core_architecture.md - name: Generate regeneration report
```

```
run: docschain report --format markdown >>
$GITHUB_STEP_SUMMARY
```

10.12 helixqa — AI-Driven QA Orchestration

Used by: CI/CD pipeline **Module path:** helixqa

helixqa orchestrates automated test suites with AI-driven test case generation, flakiness detection, and coverage gap analysis.

```
yaml # .github/workflows/helixqa.yml name: HelixQA on:
[push, pull_request] jobs: qa: runs-on: ubuntu-latest steps:
- uses: actions/checkout@v4 - name: Run helixqa run: |
helixqa run \ --suite integration \ --suite unit \ --
coverage-threshold 80 \ --flakiness-threshold 0.05 \ --ai-
suggest-missing-tests \ --report-format github
```

Integration points: - **Unit tests:** helixqa discovers all `*_test.go` files, runs them, reports coverage per package. - **Integration tests:** helixqa spins up a test Kubernetes namespace with all services and runs end-to-end test suites. - **AI test suggestion:** helixqa analyses changed code paths and suggests test cases for uncovered paths. - **Flakiness detection:** tracks test run history in its own database and flags tests with failure rate > 5%.

10.13 helixtrack.ru/core

Used by: HelixTrack Bridge Service **Module path:**

helixtrack.ru/core

```
``go // Package: helixterm.io/services/helixtrack-bridge // File:
internal/client/helixtrack_client.go
```

```
import "helixtrack.ru/core/pkg/client"
```

```
htClient := client.New(client.Config{ BaseURL: "https://
helixtrack.ru/api", OAuth2Client: helixtrackOAuth2, Timeout: 30
* time.Second, RetryAttempts: 3, })
```

```
// List issues for a project issues, err := htClient.Issues.List(ctx,
client.IssueFilter{ ProjectID: projectID, Status:
[]string{"in_progress", "open"}, })
```

```
// Link a terminal session event to an issue err =
htClient.Issues.AddEvent(ctx, issueID, client.IssueEvent{ Type:
"terminal_session", SessionID: sessionID, UserID: userID,
Summary: "Deployment via HelixTerminator", Data:
map[string]any{"host": hostname, "commands":
commandCount}, }) ``
```

10.14 HelixConstitution Compliance in CI/CD

Per §11.4.75 (mechanical enforcement without exception):

```
``yaml # .github/workflows/constitution.yml name:
HelixConstitution Compliance on: [push, pull_request] jobs:
constitution-check: runs-on: ubuntu-latest steps: - uses: actions/
checkout@v4 with: submodules: recursive

# Verify constitution submodule is present and at pinned
revision - name: Verify constitution submodule run: bash
constitution/meta_test_inheritance.sh

# Anti-bluff audit (§11.4) - name: Anti-bluff audit run: bash
constitution/scripts/audit_antibluff.sh

# Containers mandate (§11.4.76) - name: Verify containers
submodule usage run: | # Check that no service imports raw
docker/kubernetes SDK without going through
digital.vasic.containers grep -r "docker.NewClientWithOpts |
kubernetes.NewForConfig" --include="*.go" .
| grep -v "digital.vasic.containers"
| grep -v "_test.go"
&& echo "ERROR: Direct Docker/K8s SDK usage without
digital.vasic.containers" && exit 1
| | echo "OK: All container interactions use
digital.vasic.containers"

# Submodule catalogue consistency (§11.4.74) - name:
Submodule catalogue check run: | go run helixterm.io/tools/
submodule-check ./...

# Docs chain validation (§11.4.73) - name: Docs chain validation
run: docschain validate --config docs/chain/registry.go

# Push to all four upstreams (§11.4 multi-upstream) - name:
Verify push configuration run: bash constitution/
install_upstreams.sh --verify-only ``
```
